

Comparative Brain-Wide Mapping of Isoflurane and Ketamine-Activated Nuclei and Functional Networks

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Abstract

Ketamine (KET) and isoflurane (ISO) are two widely used general anesthetics, yet their distinct and shared neurophysiological mechanisms remain elusive. In this study, we conducted a comparative analysis of KET and ISO effects on c-Fos expression across the brain, utilizing principal component analysis (PCA) and c-Fos-based functional network analysis to evaluate the responses of individual brain regions to each anesthetic. Our findings demonstrate that KET significantly activates cortical and subcortical arousal-promoting nuclei, with the somatosensory cortex (SS) serving as a hub node, corroborating the top-down general anesthesia theory for dissociative anesthesia. In contrast, ISO activates the nuclei in the hypothalamus and brainstem, with the locus coeruleus (LC) as a hub node, implying a bottom-up mechanism for anesthetic-induced unconsciousness. Notably, the coactivation of sleep-wakefulness regulation, analgesia-related, neuroendocrine-related nuclei (e.g., prelimbic area (PL) and infralimbic areas (ILA), and the anterior paraventricular nucleus (aPVT), Edinger-Westphal nucleus (EW), locus coeruleus (LC), parabrachial nucleus (PB), solitary tract nucleus (NTS)) by both anesthetics underscores shared features such as unconsciousness, analgesia, and autonomic regulation, irrespective of their specific molecular targets. In conclusion, our results emphasize the distinct actions of KET and ISO while also uncovering the commonly activated brain regions, thus contributing to the advancement of our understanding of the mechanisms underlying general anesthesia.

eLife assessment

This potentially **important** study used single-cell whole-brain imaging of the immediate early gene Fos to identify the brain areas recruited by two anesthetics, ketamine and isoflurane. The utilization of a custom software package to align and analyze brain images for c-Fos positive cells stands out as an impressive component of the approach. The results suggest these anesthetics might induce anesthesia via different brain regions and pathways, and raw fos showed shared and distinct activation patterns after ketamine- v. isoflurane- vs. based anesthesia. However, the support for the primary conclusions is **incomplete** owing largely to concerns with the data transformation. The results could also be influenced by differences in route of administration between the drugs and depth of anesthesia. With these issues addressed, this paper would be of interest to preclinical and clinical scientists working with anesthetic and dissociative drugs. [\[1\]](#)

Introduction

Despite considerable investigation into the molecular targets, neural circuits, and functional connectivity associated with various anesthetics, our comprehension of their effects on overall brain activity continues to be limited and incomplete [\[1\]](#). At the molecular level, ketamine (KET) and isoflurane (ISO) interact with N-methyl-D-aspartate (NMDA) and gamma-aminobutyric acid type A (GABA_A) receptors, respectively, modulating neuronal excitability and ultimately leading to a loss of consciousness [\[2\]](#). In systems neuroscience, the neural mechanisms of anesthetic induced unconsciousness involve both top-down and bottom-up processes [\[3, 4\]](#). As evidenced by in vivo electrophysiology or functional magnetic resonance imaging (fMRI) studies, the top-down paradigm illustrates that anesthetics induce unconsciousness by disrupting corticocortical and corticothalamic circuits responsible for neural information integration, while peripheral sensory information can still be conveyed to the primary sensory cortex [\[5, 6\]](#). The bottom-up approach, exemplified by ISO, involves the activation of sleep-promoting nuclei like ventral lateral preoptic nucleus (VLPO) and inhibition of arousal centers in the brainstem and diencephalon, supporting the shared circuits of sleep and anesthesia [\[7, 8\]](#). However, the limited spatial resolution of fMRI studies and the inability of EEG to capture specific brainstem nuclei hinder the acquisition of comprehensive whole-brain information. Although a substantial body of knowledge has been amassed, our understanding of the reciprocal responses among different brain regions during general anesthesia remains relatively sparse and fragmented. To bridge these gaps, further investigation using advanced techniques that can capture the whole-brain dynamics is needed to elucidate the complex interactions and shared mechanisms between various anesthetics.

Neuronal extracellular stimulation typically results in the elevation of adenosine 3',5'-cyclic monophosphate (cAMP) levels and calcium influx, ultimately leading to the upregulation of immediate early genes (IEGs) such as c-fos [\[9, 10\]](#). The translation product of c-fos, c-Fos protein, offers single-cell spatial resolution and has been utilized as a biomarker to identify anesthetic-activated brain regions [\[11\]](#). Previous investigations of c-Fos expression throughout the brain demonstrated that GABAergic agents inhibited cortical activity while concurrently activating subcortical brain regions, including the VLPO, median preoptic nucleus (MnPO), lateral septal nucleus (LS), Edinger-Westphal nucleus (EW), and locus coeruleus (LC) [\[12–15\]](#). In contrast, KET was shown to provoke wake-like c-Fos expression and intense augmentation of c-Fos expression in various brain regions at clinical dosages (75–100 mg/kg) [\[13\]](#). It is important to note that these experiments administered KET at lights-on and GABA_A receptor agonists at lights-

off, potentially introducing circadian influences for direct comparison of ISO and KET. Moreover, it has been revealed that state of general anesthesia is not determined by activity in individual brain areas, but emerges as a global change within the brain. This change involves the activation of lateral habenular nucleus (LHb), VLPO, supraoptic nucleus (SON), and central amygdaloid nucleus (CeA), which are essential for anesthetics induced sedation, unconsciousness, or analgesia [7, 16–18]. However, brain-wide mapping and comparison of distinct anesthetic (KET and ISO) activated nuclei at a cellular level have not been fully elucidated.

In this study, we examined the distribution of nuclei activated by ISO and KET in 987 brain regions using immunochemical labeling and a customized MATLAB software package, which facilitated signal detection and registration to the Allen Mouse Brain Atlas reference [19]. We compared whole-brain c-Fos expression induced by KET and ISO through principal component analysis (PCA) and calculated inter-regional correlations by determining the covariance across subjects for each brain region pair. We then extracted significantly positively correlated brain regions to construct functional networks and performed graph theory-based network analyses to identify hub nodes. Our results uncovered distinct yet overlapping whole-brain activation patterns for KET and ISO.

Results

A comparison of the activation patterns of c-Fos in 53 brain areas in response to ISO and KET

To examine the pattern of c-Fos expression throughout the brain, 1.5% ISO was continuously ventilated, or 100 mg/kg KET was administered 90 minutes before harvesting (Figure 1A). Concurrently, to quantitatively assess brain states across the four groups, we utilized electroencephalography (EEG) and electromyography (EMG) recordings (Figure 1—figure supplement 1). The depth of anesthesia was evaluated based on the power ratio of the EEG within the delta (0.5–4 Hz) and theta (6–10 Hz) frequency bands, combined with the changes in the EMG [20]. Our findings showed that ISO and KET notably elevated EEG power ratio while reducing EMG power, suggesting both drugs induced a loss of consciousness. The raw images of brain slices were aligned to the Allen Mouse Brain Atlas (Figure 1B). Based on the standard mouse atlas available at <http://atlas.brain-map.org/>, the mouse brain was segmented into nine hierarchical levels, totaling 987 regions. The primary level consists of grey matter, the secondary of the cerebrum, brainstem, and cerebellum, and the tertiary includes regions like the cerebral cortex and cerebellar nuclei, among others. Some regions extend to the 8th and 9th levels. Our initial statistical comparison was conducted on the fifth level, which is composed of 53 subregions, as detailed in Supplementary Table 1 (Figure 1B) [21]. To differentiate c-Fos expression patterns across groups, we calculated the c-Fos density for each brain region, divided it by the overall brain c-Fos density, and then applied a logarithmic transformation. The transformed values were used for principal component analysis (PCA) (Figure 2A). The KET and ISO groups were distinctly delineated by the top two principal components. Notably, the KET-induced c-Fos expression manifested a positive association with principal component 1 (PC1), clustering around +2.5, which represented 30.72% of the total variance. In contrast, it exhibited a negative correlation with principal component 2 (PC2), clustering near -2.0, which contributed to 25.89% of the variance. On the other hand, ISO displayed a positive correlation with PC2, clustering at approximately +2.5, while its association with PC1 was neutral, near 0. Figure 2B illustrates the component coefficients of PC1 and PC2 for each brain region. Dots represent the top 25% of brain regions based on the absolute values of their component coefficients, signifying the most influential regions. KET, predominantly positively correlated with PC1, is associated with the activation of cortical regions (as evidenced by positive PC1 coefficients in MOB, AON, MO, ACA, and ORB) and the inhibition of subcortical areas (indicated by negative coefficients). On the other hand, ISO, with its notable association with PC2, showed top 25% positive coefficients in PALc, LSX of CNU, cortical ILA, and hypothalamic PVZ, while its negative coefficients were found primarily

in cortical regions. These observations underscore that KET and ISO exhibit distinct patterns of brain region activation. KET displays cortical activation and subcortical inhibition, whereas ISO shows a contrasting preference, activating the cerebral nucleus (CNU) and the hypothalamus while inhibiting cortical areas. To reduce inter-individual variability, we normalized c-Fos⁺ cell counts in each brain area by dividing them by the total c-Fos⁺ cells in the entire brain (**Figure 2C**), Subsequently, we directly compared the differences between ISO and KET with their respective control groups. In line with PCA results, KET significantly activated the isocortex (16.55 ± 2.31 vs. 40.13 ± 1.97 , $P < 0.001$) and reduced the proportion of c-Fos⁺ cells in the hypothalamus (14.88 ± 2.18 vs. 4.03 ± 0.57 , $P = 0.001$) and midbrain (21.87 ± 2.74 vs. 6.48 ± 1.06 , $P = 0.005$) compared to the saline group. In contrast, the proportion of c-Fos⁺ cells induced by ISO were mainly observed in the striatum and hypothalamus compared to the home cage group, specifically in the periventricular zone (PVZ) (2.330 ± 0.759 vs. 6.63 ± 0.84 , $P = 0.015$) and lateral zone (LZ) (2.82 ± 0.3 vs. 5.59 ± 0.75 , $P = 0.002$) of the hypothalamus, as well as STRv (0.59 ± 0.10 vs. 2.77 ± 0.40 , $P < 0.002$) and LSX (1.93 ± 0.59 vs. 5.11 ± 0.41 , $P = 0.004$) of the striatum (**Figure 2C** and Table 1). In summary, our findings reveal that KET predominantly affects cortical regions, whereas isoflurane mainly targets subcortical areas, specifically the striatum, and hypothalamus. These results demonstrate distinct patterns of brain region activation for each anesthetic agent.

Similarities and differences in ISO and KET activated c-Fos brain areas

To elucidate the c-Fos expression patterns induced by ISO and KET more clearly, we refined our focus from the initial 53 brain regions to a more granular examination of subregions, culminating in a comprehensive assessment of 201 distinct areas. We conducted a PCA on the logarithmic ratio of c-Fos density across these regions to discern variance among the groups (**Figure 3**). The KET group predominantly clustered around +7 on PC1 and approximately 0 on PC2, whereas the ISO group clustered around +4 on PC2 and -3 on PC1. KET displayed a strong positive correlation with PC1, accounting for 28.97% of the variance. In contrast, ISO negatively correlated with PC1 and positively with PC2, which accounted for 15.54% of the variance (**Figure 3A**). These findings underscore the distinct features between the ISO and KET groups, with the top two principal components effectively separating them. Further analysis of PC coefficients in each brain region showed (**Figure 3B**) that KET, positively associated with PC1, mainly activates the cerebral cortex (CTX) areas including the visceral area (VISC), claustrum (CLA), dorsal peduncular area (DP), orbital area (ORB), and temporal association areas (TEA), while inhibiting the hypothalamus, midbrain, and hindbrain. In contrast, ISO is negatively correlated with PC1, suggesting that the cortical areas activated by KET are inhibited by ISO. Additionally, ISO has a positive association with PC2 and is predominantly related to the activation of hypothalamic regions, such as the ventrolateral preoptic nucleus (VLPO), suprachiasmatic nucleus (SCH), tuberal nucleus (TU), medial preoptic area (MPO), and supraoptic nucleus (SON), as well as certain nuclei in the central nucleus (CNU). KET primarily activates regions within the cerebral cortex (CTX) and inhibits areas within the hypothalamus, midbrain, and hindbrain, aligning with the top-down mechanism suggesting anesthetic agents influence cortical and thalamocortical circuits [4]. Conversely, ISO activates hypothalamic areas while suppressing the cortex, midbrain, and hindbrain, consistent with the bottom-up mechanism. **Figure 3C** summarizes these distinctive responses, implying that anesthetic agents like KET and ISO modulate consciousness by targeting particular brain regions and neural circuits.

Identification of brain regions activated by KET

We further employed quantitative methods to identify the brain regions activated by KET across the entire brain. Our findings concur with previous studies: numerous cortical regions associated with somatosensory, auditory, visual, and movement were activated (**Figure 4A and C**, **Supplementary Figure 1**, **Supplementary Table 2**). Additionally, we identified several

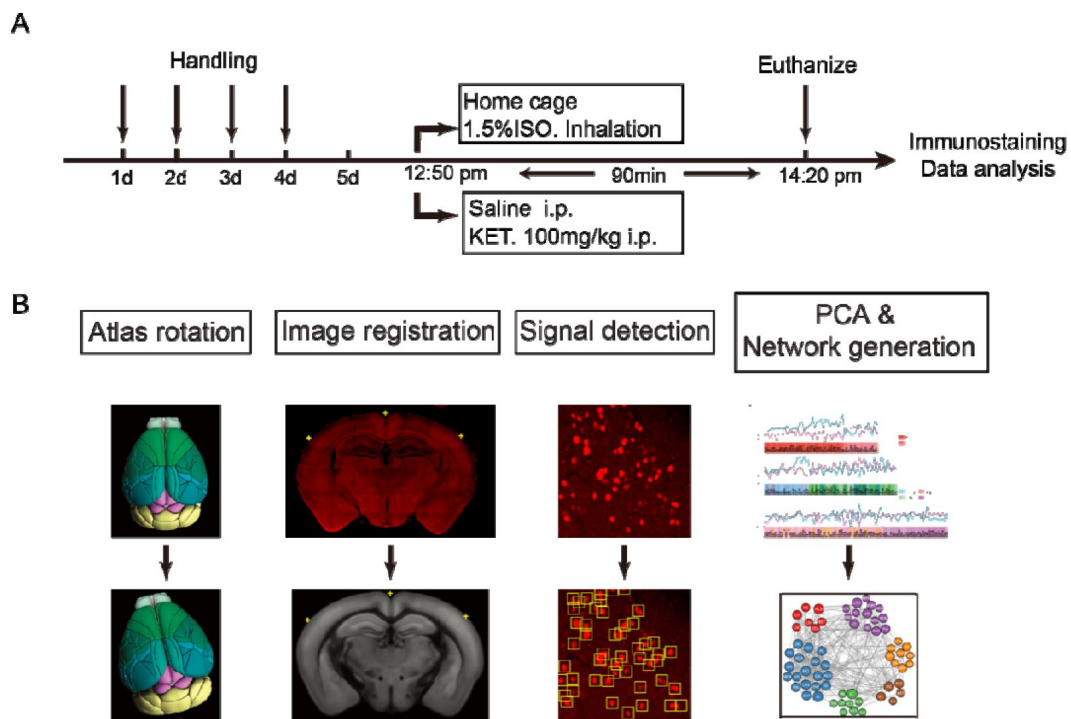


Figure 1.

Brain-wide quantification of c-Fos expression.

(A) Schematic representation of the habituation protocol typically used to acclimate mice. After being exposed to anesthetics for 90 minutes, the mice were euthanized. (B) Steps for data processing. Example of brain section registration to a corresponding coronal section from the Allen Brain Atlas. For Atlas rotation, the Allen reference atlas was rotated to mimic the slice angle of the experimental brain. Image registration maps the original coronal image (upper panel) to the corresponding Allen mouse brain atlas (lower panel). The registration module applies several geometric transformations (translation, rotation, and scaling) to optimize the matching of the original image to the anatomical structures. Fluorescence signals were detected from the original image (upper panel), and once detected, they were projected onto the Allen Mouse Brain Atlas for quantification and network analysis by means of the detected signals labeled with yellow boxes.

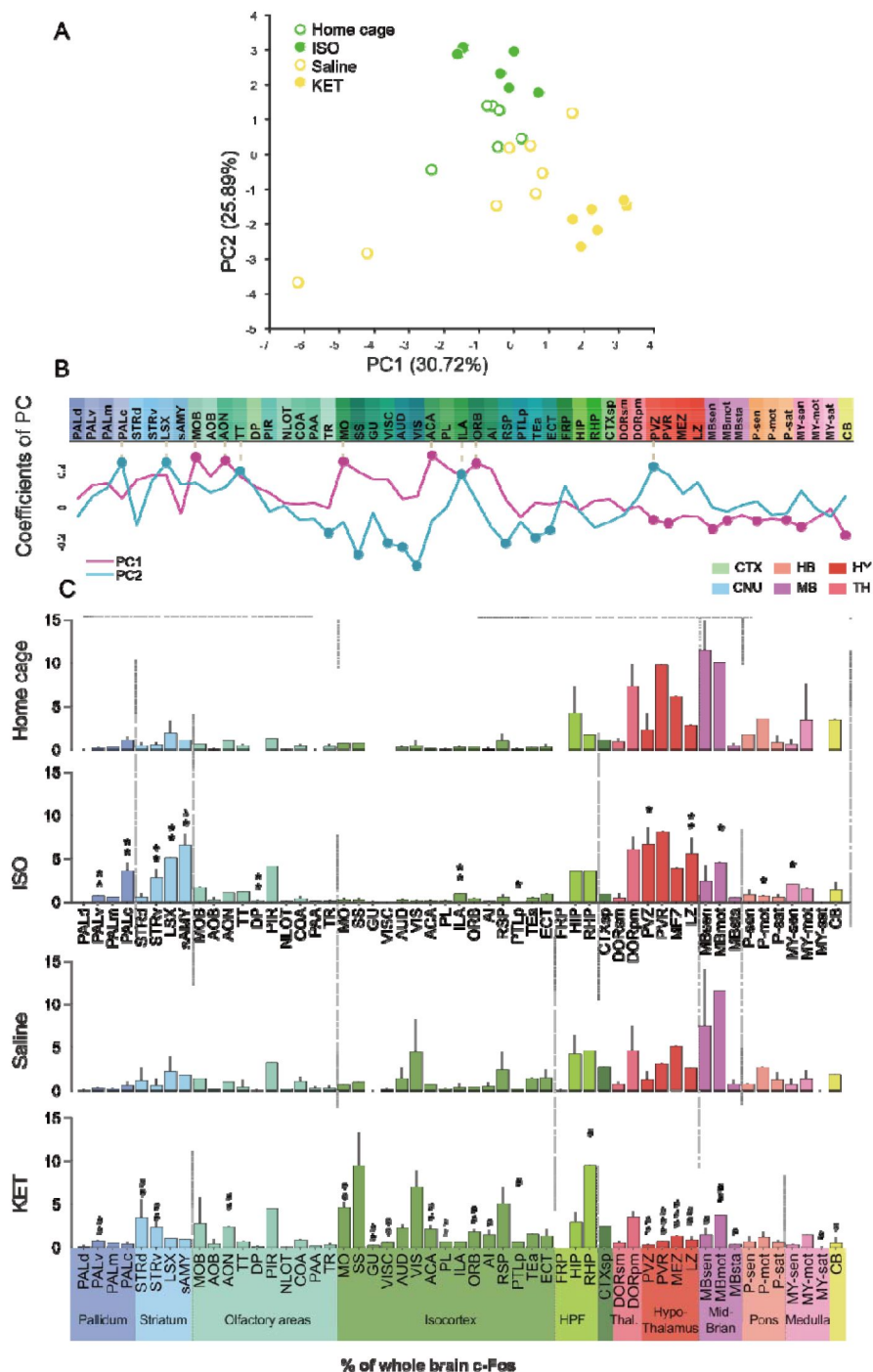


Figure 2.

Whole-brain distributions of c-Fos⁺ cells induced by ISO, KET, and control conditions.

(A) Scatter plot of four conditions in the space spanned by the first two principal components (PC1 versus PC2) of c-Fos density from 53 brain regions. (B) Line plot of coefficients of PC1 and PC2. Dots represent regions with absolute values larger than 75 percentiles. (C) The normalized c-Fos⁺ cells in 53 brain areas (Home cage, n = 6; ISO, n = 6 mice; Saline, n = 8; KET, n = 6). Brain areas are grouped into 12 generalized, color-coded brain structures. Abbreviations of the 53 brain areas and percentages of c-Fos⁺ cells are listed in [supplementary table 1](#). Error bar, mean ± SEM. Mann-Whitney tests were corrected using the false discovery rate approach with a Q-value set at 0.05. Home cage vs ISO: *P<0.05, **P<0.01, ***P<0.001. Saline vs KET: #, P<0.05, ##, P<0.01, ###, P<0.001.

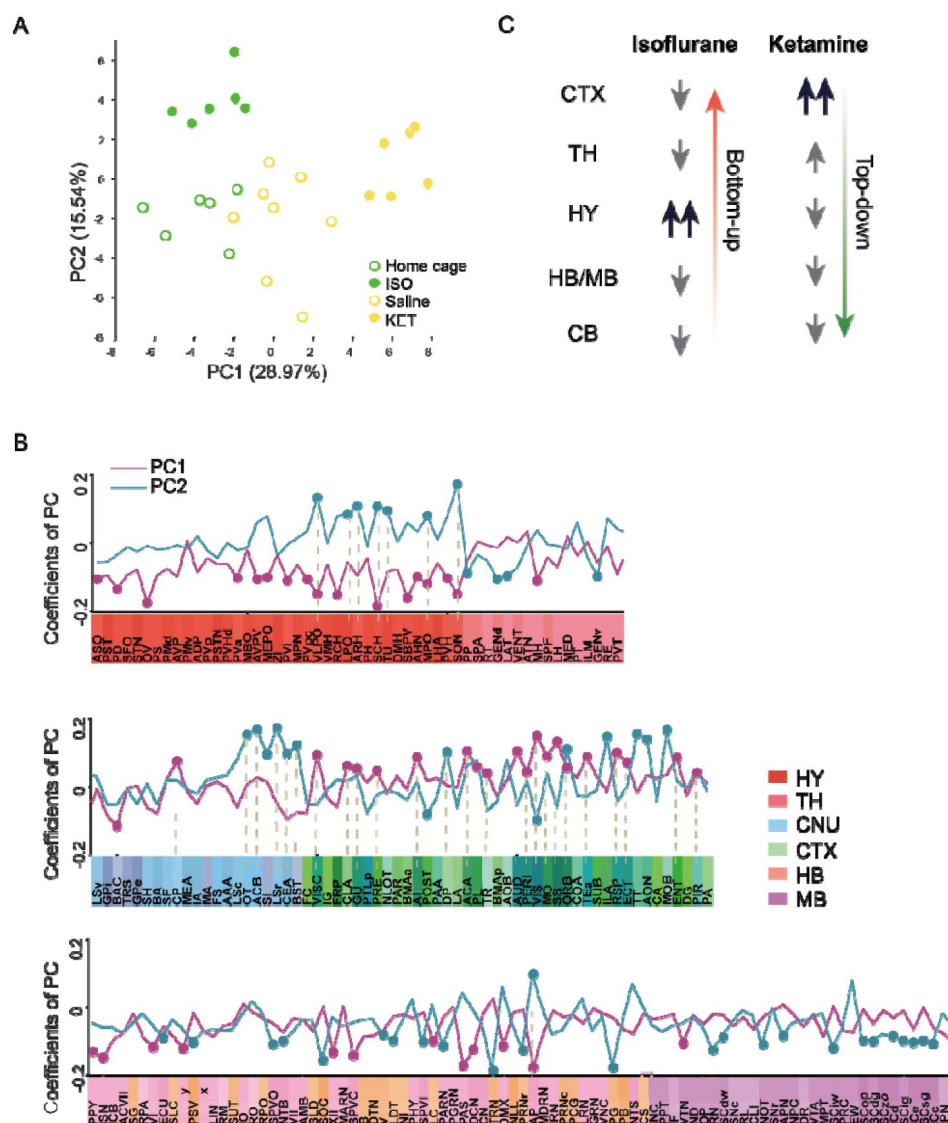


Figure 3.

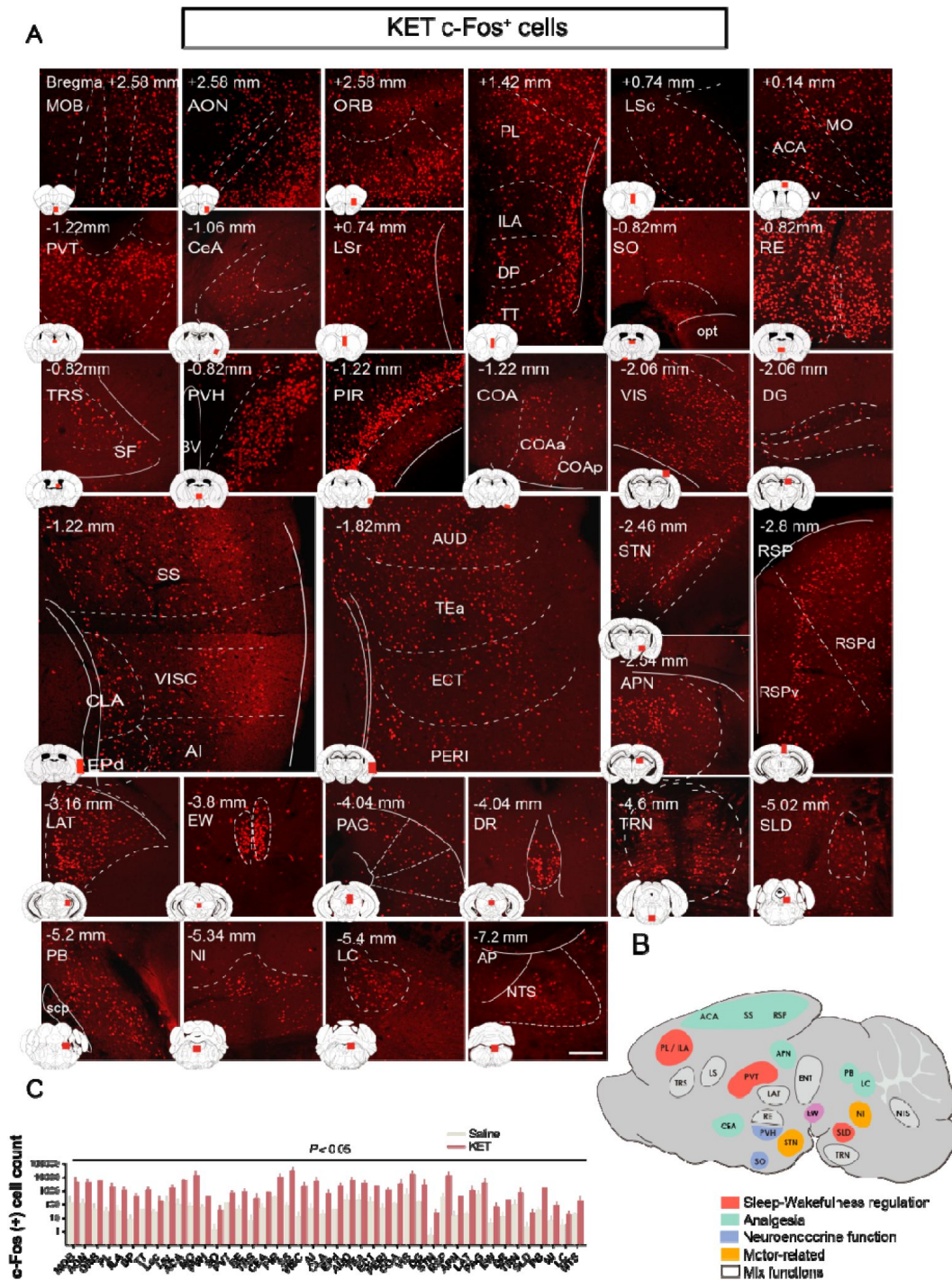
Similarities and differences in ISO and KET activated c-Fos brain areas.

(A) Scatter plot of four conditions in the space spanned by the first two principal components (PC1 versus PC2) of normalized c-Fos density from 201 brain regions. (B) Line plot of coefficients of PC1 and PC2. Dots represent regions with absolute values larger than 75 percentiles. (C) Activation patterns of ISO and KET in different parts of brain regions were summarized. ISO was predominantly characterized by increased activation of hypothalamic and cerebral nuclei while reducing activity in the cortex and midbrain, indicating that ISO primarily exerts its effects through a bottom-up mechanism. On the other hand, KET primarily activated the cortex and inhibited hypothalamus and midbrain activity, suggesting that KET regulates consciousness through a top-down mechanism.

innovative observations that enrich the current understanding in this field. To provide a clearer overview of our findings, we categorized the activated brain areas into different functional groups: (1) Arousal and REM sleep regulation: Several nuclei associated with arousal were activated, including the prelimbic area (PL), infralimbic cortex (ILA), paraventricular nucleus of the thalamus (PVT), dorsal raphe (DR), and sublaterodorsal nucleus (SLD). Prior evidence indicates that the PL/ILA and PVT regions play a role in regulating arousal in both cortical and subcortical areas [22, 23]. Additionally, dopaminergic neurons within the DR have been identified as vital components of wake-promoting pathways [24]. In the context of REM sleep, the sublaterodorsal nucleus (SLD) stands out for its role in stabilization [25]. (2) Pain modulation, KET significantly activated pain-related areas such as the anterior cingulate cortex (ACA) in the cortex [26], the anterior pretectal nucleus (APN) in the thalamus, which is known to be involved in managing chronic pain [27], and the anterior periaqueductal gray (PAG) region and medial prefrontal cortex (mPFC), both part of the endogenous pain inhibitory pathway [28, 29]. Additionally, the activation of the locus coeruleus (LC) in the midbrain may contribute to KET's analgesic effects [30]. (3) Neuroendocrine regulation: The paraventricular hypothalamic nucleus (PVH) and supraoptic nucleus (SON), which are neuroendocrine-related regions, were also activated [18, 31]. (4) Movement: Subcortical nuclei associated with movement, such as the subthalamic nucleus (STN) and nucleus incertus (NI), were prominently activated by KET administration [32, 33]. (5) Connectivity: We observed significant activation of the nucleus reuniens (Re) located in the thalamus, which receives substantial afferent input from limbic structures and serves as a connector linking the hippocampus to the medial prefrontal cortex [34]. While our functional categorization offers a simplified overview, it's worth noting that each activated brain region may have multifaceted roles beyond this classification. In summary, our study identified extensive activation of cortical and subcortical nuclei during KET anesthesia, encompassing regions related to arousal, pain modulation, neuroendocrine regulation, movement, and connectivity.

Identification of brain regions activated by ISO

In our study, we aimed to identify brain regions activated by ISO and compare these with those activated by KET. We began by summarizing previously reported ISO activated nuclei, including PIR, LSd/LSv, CeA in the cortex and striatum, and VLPO, MnPO, EW, NTS, LC, ventral group of the dorsal thalamus (VeN), and area postrema (AP) in the hypothalamus and midbrain (**Supplementary Table 3**). We subsequently conducted a comprehensive MATLAB-based analysis of c-Fos expression throughout the entire brain, uncovering previously undetected activated nuclei (**Figure 5A**, **Supplementary Figure 2**). Newly identified activated nuclei in the CTX and the CNU included PL/ILA and ENT, aPVT in the thalamus, TU, ARH, PVi, and PVH in the hypothalamus, and PB in the hindbrain. All nuclei activated by ISO in this study were functionally classified and depicted in **Figure 5C**. Our results confirmed the activation of several brain regions involved in arousal-related nuclei, such as the prelimbic area (PL) and infralimbic areas (ILA), and the paraventricular nucleus (PVT) [22, 23]. Additionally, we observed activation in previously reported analgesia-related nuclei, including CeA and LC, as well as the parabrachial nucleus (PB) [18, 30, 35]. We also found activation in neuroendocrine function-related nuclei of the hypothalamus, such as TU, PVi, ARH, PVH, and SON. Moreover, we identified activations related to ISO induced side effects, such as in the piriform cortex (PIR) [36] and ENT [37], which may be stimulated by ISO odor, and the solitary tract nucleus (NTS), potentially responsible for ISO induced vomiting [38]. The only activated nucleus in the midbrain was the Edinger-Westphal nucleus (EW). Recent research has found that sevoflurane activates EW and is involved in sleep induction and maintenance of anesthesia, suggesting its crucial role in general anesthesia [39]. By comparing the ISO and KET induced c-Fos expression, we summarized the brain regions activated by both anesthetics in **Figure 5D**. Despite variations in molecular targets, the coactivation of regions such as PL/ILA, aPVT, CeA, PVH, SON, EW, PB, LC, and NTS by both ISO and KET suggests an overlapping neuronal circuitry that influences sleep-wake regulation, analgesia, and neuroendocrine functions. This shared neural circuitry may potentially offer a common mechanism across the two anesthetics for the maintenance of general anesthesia.



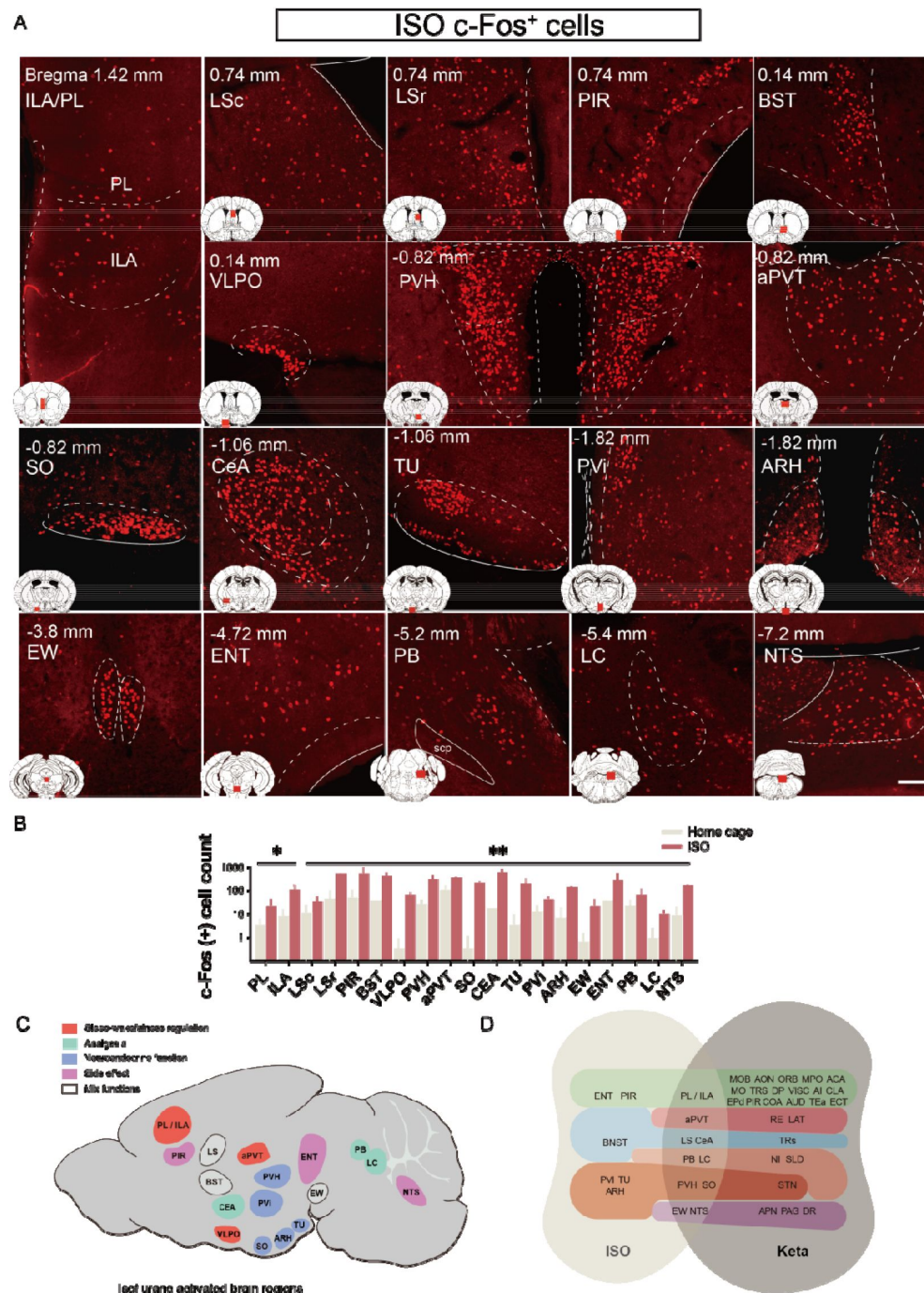


Figure 5.

c-Fos expression in distinct brain regions after exposure to ISO.

(A) Representative of brain regions with statistical differences c-Fos⁺ cells between the ISO group and home cage mice. Scale bar, 200 μ m. (B) Representation of cell counts in brain regions exhibiting significant statistical differences between home cage and ISO. Data are shown as mean $\pm$ SEM. P values < 0.05. (C) Schematic cross-section of the mouse brain showing activated brain regions by ISO. Different colors indicate various functionally relevant nuclei. Red signifies nuclei involved in sleep-wake regulation, blue-green in pain management, blue in neuroendocrine function, pink in side-effect management, and white denotes nuclei exhibiting mixed functionalities. (D) The Venn diagram shows brain regions that are co-activated by ISO and KET and differentially activated brain regions.

Network generation and Hub identification

Previous research has established that general anesthesia is mediated by various brain regions [7, 16–18]. c-Fos expression serves as an indicator of neuronal activity, providing a single index of activation per region per animal. By examining the covariance of c-Fos expression across animals within each group, we can infer interactions between brain regions and identify functional networks engaged during general anesthesia [40]. Consequently, we initially incorporated a total of 63 brain regions, including nuclei that were significantly activated by ISO and KET (as indicated by positive findings in **Figures 4** and **5**) and those previously reported to be associated with the mechanisms of general anesthesia and sleep-wakefulness [41]. Highly correlated activities across different brain regions are presumed to constitute components of a functional network, reflecting the complex interplay of brain areas during general anesthesia. Using c-Fos expression as a marker for neuronal activity and calculated a comprehensive set of interregional correlations for four groups. The matrices exhibited interregional correlations for the number of c-Fos-positive cells in each condition (**Figure 6A**). Network graphs were generated by extracting Pearson's coefficients > 0.82 , as well as significant positive correlations ($P < 0.05$), from these matrices to construct functional brain networks (**Figure 6B**). Compared to the control group, isoflurane (ISO) slightly decreased interconnections between regions (network density: 0.13 vs. 0.10; edges: 267 vs. 198), with no significant difference in mean interregional correlation coefficients between the ISO group and the home cage group (Fisher $Z = -0.018$, $P = 0.98$). Conversely, ketamine (KET) significantly increased the network's connectivity density (network density: 0.13 vs. 0.47; edges: 265 vs. 1008) and showed a significant increase in mean interregional correlation coefficients compared to the saline group (Fisher $Z = 3.54$, $P < 0.001$) (**Figure 6C**). These findings suggest that ISO may exert a mild inhibitory effect on functional network connectivity, whereas KET appears to enhance interregional correlation following administration (**Figure 6D**).

Hubs are nodes that occupy critical central positions within the network, enabling the network to function properly. To determine the brain regions that serve as the hubs for ISO and KET induced functional networks, we calculated each node's degree (the number of links that connect it) and betweenness centrality (the fraction of all shortest paths in the network that contain a given node). Nodes with high degree and betweenness centrality values typically exhibit many connections. We ranked each node according to its degree and betweenness centrality and extracted nodes with a rank greater than the 80th percentile in its network (**Figure 7**). Additionally, we segmented the functional network into non-overlapping modules using a spectral community detection algorithm and calculated the within-modal degree Z-score (the degree of nodes within a module, which indicates within-module connectivity) as well as participation coefficients (the distribution of the edges of a node between other modules, which indicates within-module connectivity) [42]. Nodes with relatively high values for both parameters were considered relatively significant within and outside the network module. We classified nodes with participation coefficients > 0.4 and Z scores > 1 as connector hubs [43]. The LC, exhibiting elevated degree and betweenness centrality as well as relatively higher Z-scores and participation coefficients, may play a pivotal role in mediating isoflurane-induced general anesthesia, given its known involvement in arousal, attention, and analgesia. In contrast, the somatosensory cortex (SS) functions as a connector hub in the KET group, indicating its integrative and coordinating role in ketamine-induced dissociative anesthesia (**Figure 7D**). The LSc served as the central hub in the saline group (**Figure 7C**). No significant findings were observed in the within-modal degree Z-score and participation coefficient analyses for the home cage group. However, the analysis did reveal a relatively high degree and betweenness centrality of APNs (**Figure 7A**), suggesting that the within-module connections are relatively independent, or that the pathways of inter-module information transmission do not depend on specific connector nodes.

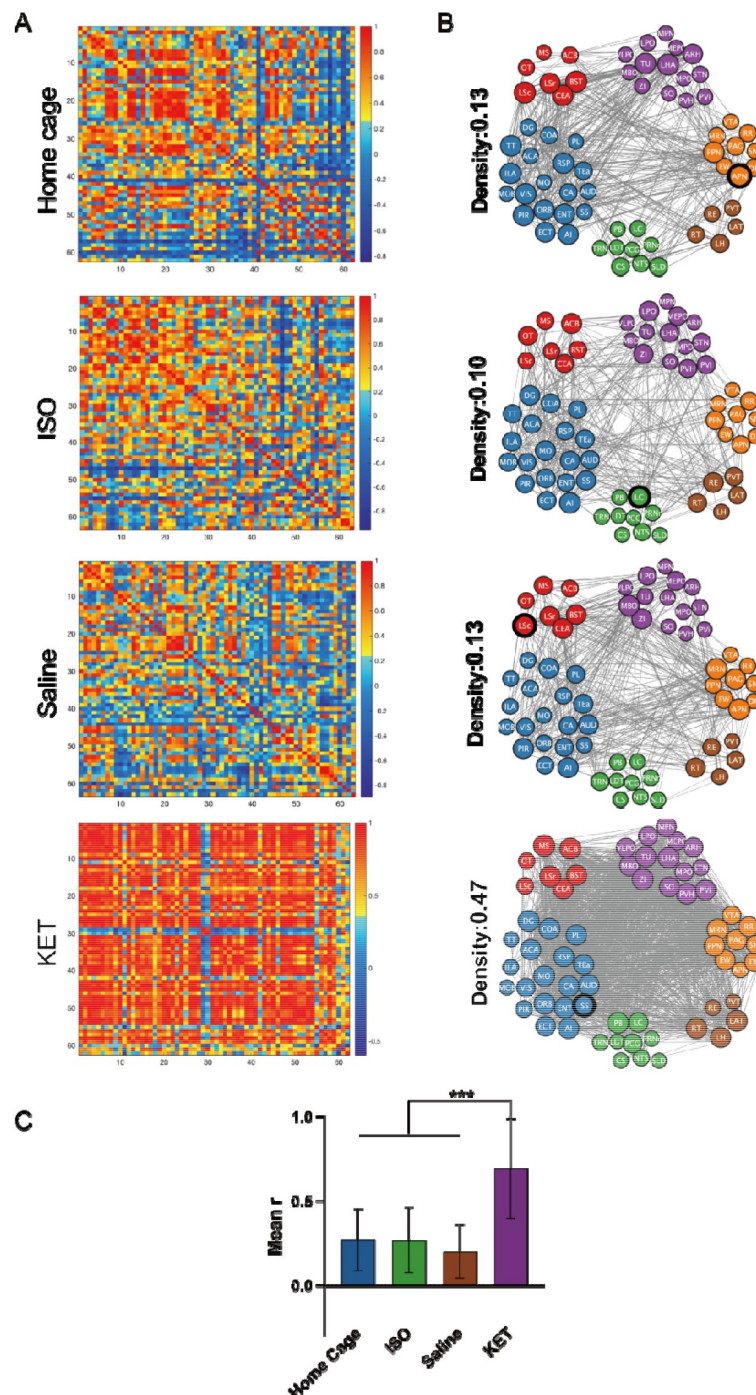


Figure 6.

Generation of anesthetics-induced networks and identification of hub regions.

(A) Matrices showing interregional correlations for c-Fos expression at the home cage, ISO, saline, and KET. Colors reflect correlation strength based on Pearson's r (color bar, right). Axes are numbered and correspond to brain regions listed in [Supplementary Table 5](#). (B) Network graphs were generated by significant positive correlations ($P < 0.05$), as well as Pearson's $r > 0.82$. The widths of the edges are proportional to the strength of the correlations and the size of the nodes is proportional to the degree of the node. Colors represent the major brain division (red, cerebral nuclei; purple, hypothalamus; orange, midbrain; brown, thalamus; blue, thalamus; green, the cerebral cortex). Network densities were noted in the left. (C) Mean r was calculated from interregional correlation coefficients. Data were analyzed using one-way ANOVA, followed by Tukey's post hoc test. *** $P < 0.001$.

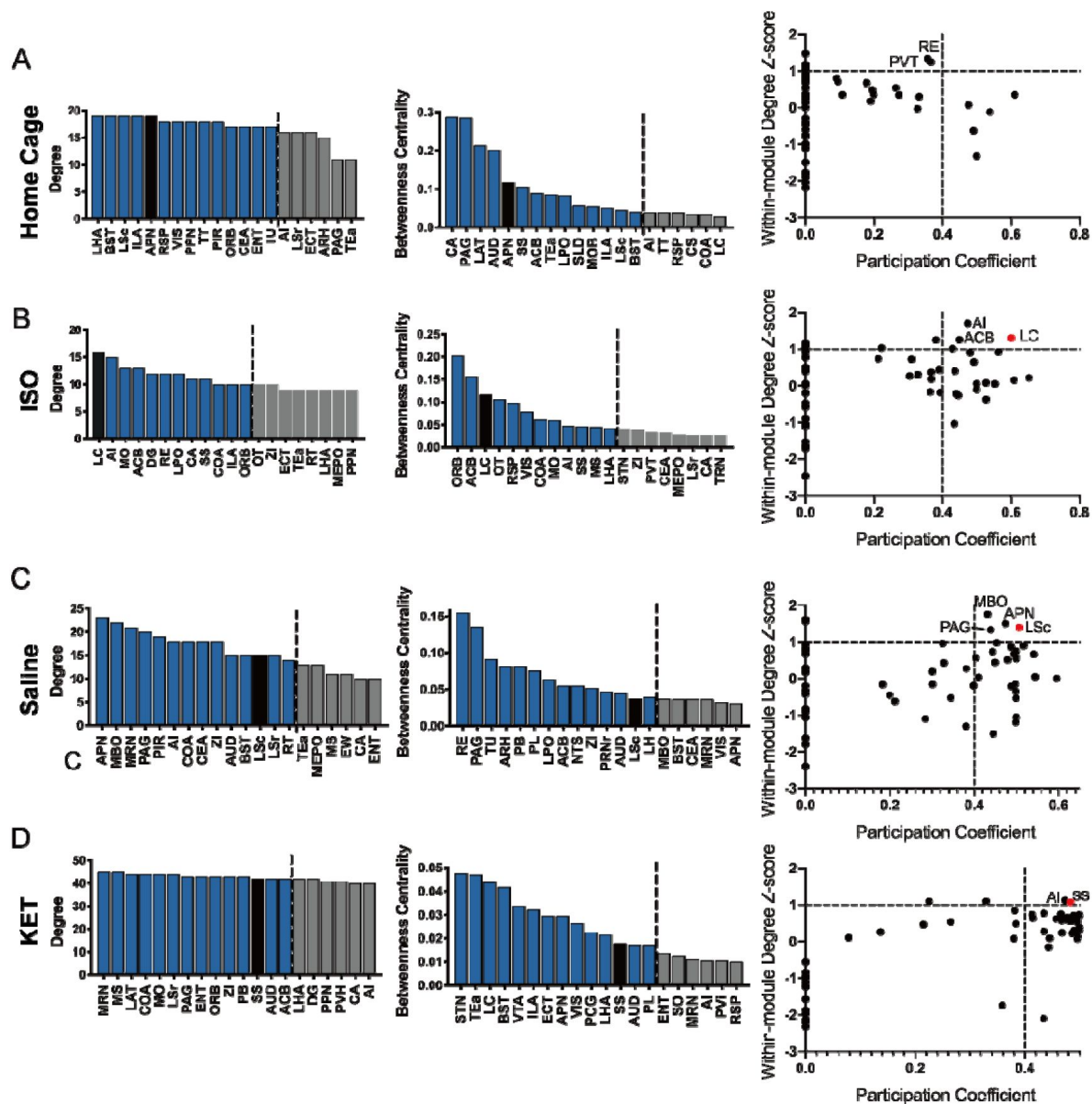


Figure 7.

Identification of hub regions in the home cage (A), ISO (B), NS (C), and KET (D) groups.

Each brain region is ranked in descending order based on degree and betweenness centrality, respectively. The regions to the left of the dotted line represent the top 20% in our network. Black columns highlight brain regions that rank similarly in both degree and betweenness centrality. The within-module Z-scores and participation coefficients were calculated for each brain region in the networks. Brain regions characterized by a participation coefficient greater than 0.4 and a within-module degree Z-score greater than 1 are defined as hubs and are denoted by red dots.

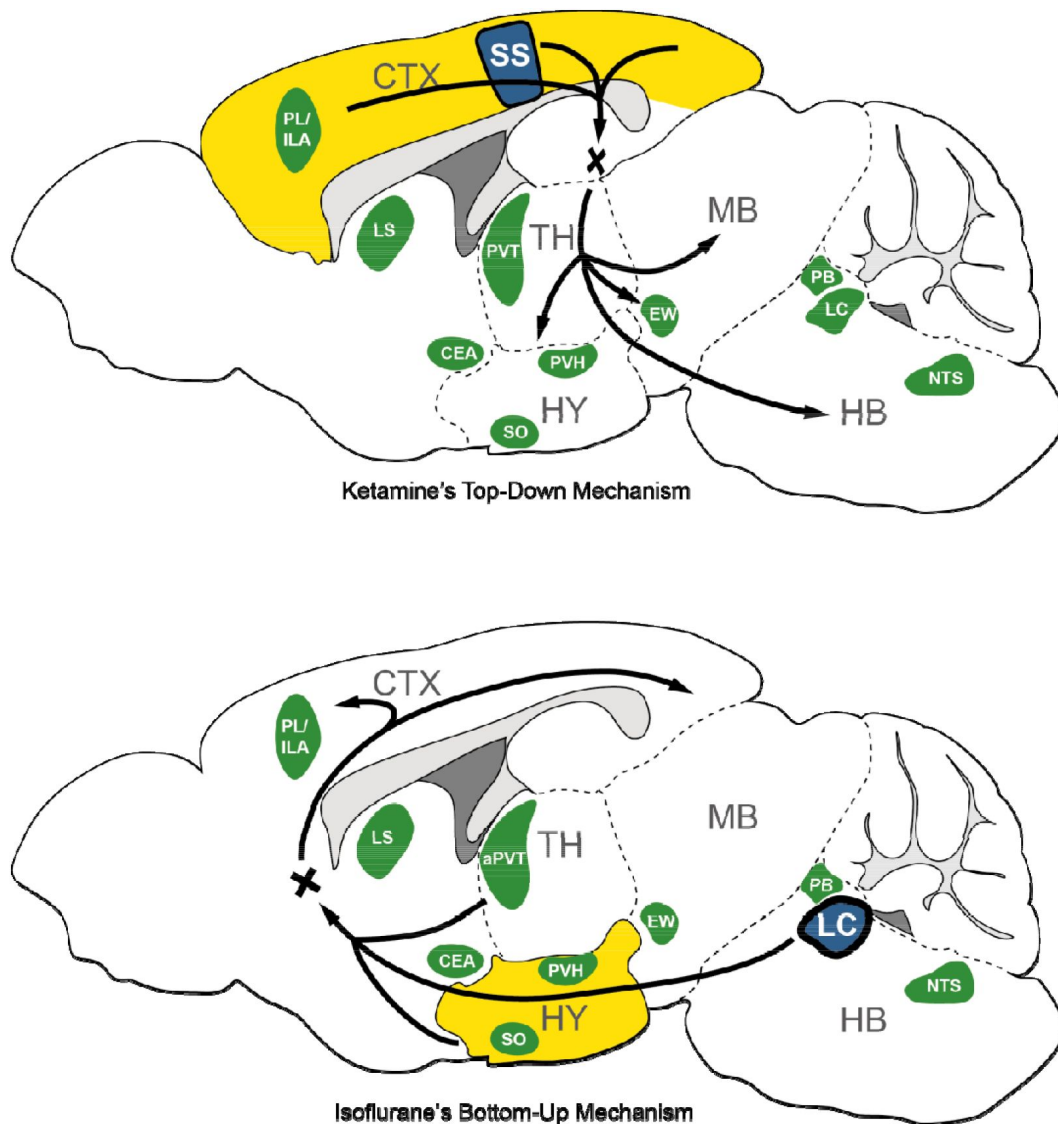


Figure 8.

The possible framework for ketamine and isoflurane induced unconsciousness.

The distinct pathways of ketamine and isoflurane induced unconsciousness can be explained by two contrasting mechanisms. The “top-down” process attributes KET’s effect to widespread cortical activation (represented in yellow), with the somatosensory cortex (SS) acting as the central node in the functional network (depicted in blue) and relative inhibition of hypothalamic sleep-promoting nuclei. Conversely, the “bottom-up” approach posits that isoflurane induced unconsciousness arises from the activation of hypothalamic sleep-promoting regions (indicated in yellow) and relative inhibition of cortical and thalamic nuclei, with the locus coeruleus (LC) serving as the hub node in the isoflurane induced functional network. Nuclei activated by both anesthetics are shown in green. Adapted from [3, 4, 52]. PL, prelimbic area; ILA, infralimbic areas; SON, supraoptic nucleus; PVH, paraventricular hypothalamic nucleus; TH, thalamus; HY, hypothalamus; MB, midbrain; HB, hindbrain.

In this study, we conducted a comparative analysis of the effects of two general anesthetics, isoflurane (ISO) and ketamine (KET), on c-Fos expression throughout the brain. By employing principal component analysis (PCA), we were able to thoroughly examine the responses of individual brain regions to each anesthetic agent. Our findings reveal that KET dominantly activates the cerebral cortex yet suppresses subcortical regions, reflecting a top down mechanism of action, while ISO predominantly stimulates subcortical brain regions with relative cortical inhibition, substantiating its bottom up mechanism of action [3, 4]. Further functional analysis of brain networks, based on c-Fos expression, identified the somatosensory cortex (SS) and the locus coeruleus (LC) as central nodes for KET and ISO, respectively, highlighting the crucial roles of LC and SS under ISO and KET induced unconsciousness.

Our results demonstrate that ISO activates the sleep-promoting VLPO nucleus, the aPVT-infralimbic loop [23], and broadly inhibits the cortex, supporting a bottom-up mechanism of ISO induced unconsciousness. Nonetheless, our findings also reveal the activation of arousal related nuclei, such as PB and LC, which implies that the influence of isoflurane on consciousness may not solely rely on suppressing arousal centers, but rather through a more intricate relationship than formerly recognized. Identifying cell types and their dynamic changes during anesthesia will be crucial for clarifying their role in ISO induced unconsciousness.

While c-Fos-based functional network analysis offers lower temporal resolution, its single-cell resolution across the entire brain allows for the inclusion of midbrain and hindbrain regions, supplementing previous fMRI analyses. Our observations reveal that ISO mildly inhibits network density, and through graph theoretical analysis, we identify the LC as a highly connected hub, highlighting the critical role of the brainstem in ISO induced general anesthesia. The LC performs a wide range of functions in mice, including arousal, pain modulation, attention, stress response, and neuroprotection. Studies have shown that chemical activation of the LC increases whole-brain functional connectivity, attributed to its role as the primary source of norepinephrine (NE) and its extensive influence on nearly the entire brain [44]. The significant LC activation and its central position within the functional network underlying ISO induced unconsciousness suggest that the LC plays a crucial part in maintaining and integrating the entire unconsciousness functional network, emphasizing the involvement of LC in the bottom-up paradigm of ISO induced unconsciousness.

Mashour et al. proposed that anesthesia-induced unconsciousness encompasses not only the modulation of lower-level brain activity but also top down neural processing [3, 4]. Within this top down framework, anesthetics diminish consciousness by interfering with cortical and thalamocortical circuits responsible for neural information integration. Our study discovered that KET administration substantially activated cortical and subcortical arousal-promoting nuclei while concurrently causing relative thalamic suppression, with only the RE and TRS exhibiting activation. This suggests that thalamic inhibition may lead to a reduction in thalamocortical communication, which is characterized by the inability to perceive the external environment and results in disconnection from reality. Graph theoretical analysis also identified the somatosensory cortex (SS) as the hub node of the KET induced functional network. As a critical cortical area, SS is responsible for sensory processing, motor control, and cognitive functions [45]. Previous studies have demonstrated that local KET administration to SS recapitulates the effects of systemic KET on both the switch in pyramidal cell activity and dissociative-like behavior, implying that SS may serve as a key target for KET induced dissociation [46]. Our findings that SS acts as a hub node suggest that KET may modulate brain network function by influencing connectivity between SS and other brain regions, thereby affecting the behavior and cognitive states of mice. This further supports the significance of cortical areas during KET anesthesia.

Identifying shared neural features between KET and ISO is essential for understanding anesthetic-induced unconsciousness. The coactivation of sleep-wake regulation-related regions, such as PL/ILA and aPVT, along with analgesia-related nuclei like CeA, PB, and LC, suggests a shared mechanism for sleep-wake regulation and the common pathways for pain relief. This observation provides valuable insights into the fundamental mechanisms of anesthesia-induced hypnosis and analgesia. Additionally, the coactivation of neuroendocrine-related nuclei, including PVH and SO in the hypothalamus, raises questions about the potential influence of anesthetics on hormonal release and homeostatic regulation. Other coactivated nuclei, such as EW and NTS, warrant further investigation of their roles in anesthesia. In summary, the coactivated nuclei imply a potential shared neuronal circuitry for general anesthesia, encompassing common features like unconsciousness, analgesia, and autonomic regulation, regardless of the specific molecular targets of each drug. Future research could examine coactivated brain regions by the two anesthetics or manipulate identified hub nodes to further understand the mechanisms of general anesthesia. In summary, our study reveals distinct and shared neural mechanisms underlying isoflurane and ketamine anesthesia using c-Fos staining and network analysis. Our findings support “top-down” and “bottom-up” paradigms, and the identification of hub nodes and coactivated brain regions suggests shared neurocircuitry for general anesthesia, providing insights into the mechanisms underlying anesthetic-induced unconsciousness and analgesia.

Methods

Animals

All animal experiments were conducted in accordance with the National Institutes of Health guidelines and were approved by the Chinese Academy of Sciences’ Institute of Neuroscience. Adult male wild-type (WT) mice (C57BL/6J) (8-10 weeks old, weight from 22 to 26 g) were purchased from institute-approved vendors (LingChang Experiment Animal Co., China). Mice were individually housed and maintained on a 12 h:12 h light/dark cycle (lights on at 07:00 a.m. and off at 07:00 p.m.) with food and water available ad libitum.

Drug administration

All experiments occurred between 13:00-14:30 (ZT6–ZT7.5). We adapted mice to handling and the anesthesia chamber (10×15×15 cm) for several days to minimize experimental confound-induced c-Fos induction. Adult male mice were handled for the KET group for 10 min per day with normal saline (NS) injected intraperitoneally (i.p.) for three consecutive days at 13:00. On day five, a randomly chosen mouse received an injection of Ketamine (Gutian Medicine, H35020148), and the control groups (n=8) received the same volume of saline. ISO group (n=6) mice were handled and inhaled 1.5% isoflurane (RWD Life Science, 1903715) at 13:00 on day four in the chamber. Meanwhile, the control groups (n=6) were left undisturbed in their home cages prior to sampling. We confirmed the loss of righting reflex at 5 min after anesthetics exposure. For 90 min after KET injection or ISO inhalation, mice were deeply anesthetized with 5% ISO and transcardially perfused with 35 ml 0.1 M phosphate-buffered saline (PBS) followed by 35 ml 4% paraformaldehyde (PFA). The brains were then removed and postfixed overnight with 4% PFA. Following fixation, the brains were dehydrated for 48 hours with 30% sucrose (wt/vol) in PBS. Coronal sections (50 μm) of the whole brain were cut using a cryostat (HM525 NX, Thermo Scientific) after being embedded with OCT compound (NEG-50, Thermo Scientific) and freezing.

Anesthesia depth measurement

EEG and EMG recordings were conducted as described previously^[20]. Specifically, a stainless steel electrode was positioned over the frontal cortex for EEG measurements, while an insulated EMG electrode was inserted into the neck muscles. Additionally, a reference electrode was placed above the cerebellum. The EEG/EMG data were captured using TDT system-3 amplifiers,

specifically RZ5 + RA16PA and RZ2 + PZ5 configurations. To measure the depth of anesthesia using EEG/EMG signals, a fast Fourier transform spectral analysis was employed, featuring a frequency resolution of 0.18 Hz. Data was noted every 5 seconds through a MATLAB tool and subsequently verified manually by experts. The assessment was based on EMG power and the ratio of EEG's δ power (0.5–4Hz) to θ power (6–10Hz). EEG and EMG power values within 30 minutes post-administration were normalized to a 5-minute pre-administration baseline.

Immunohistochemistry

One out of every three brain slices (100 μ m intervals) of each whole brain was washed three times with 0.1 M phosphate-buffered saline (PBS) for 10 min and permeabilized for 30 minutes at room temperature with 0.3% Triton X-100 in PBS (PBST). Slices were incubated for 2 hours at room temperature with 2% normal donkey serum (Sigma, G6767) in PBS overnight at 4°C with c-Fos primary antibodies (226003, Synaptic Systems; 1:500) diluted in PBS with 1% donkey serum. After three washes with PBST, slices were incubated with the Cy3 donkey anti-rabbit (711165152, Jackson; 1:200) secondary antibody for 2 hours at room temperature. Immunostained slices were mounted with VECTASHIELD mounting medium with DAPI and then scanned under a fluorescent microscope equipped with a 10 $\times$ objective (VS120, Olympus) or a confocal microscope with a 20 $\times$ objective (FV300, Olympus). Due to the limited scope of a single field of view, either 2 or 4 adjacent fields of view were stitched together to offer a comprehensive representation of specific brain regions, including the PL, ILA, LSc, LSr, SS, and VISC.

Quantification of c-Fos positive cells

The procedures used for c-Fos analysis were based on previous research [19]. A custom-written software package was employed for cellular quantification within brain images. The software consists of three components: atlas rotation, image registration, and signal detection.

Atlas rotation module

Utilizing the Allen Mouse Brain Atlas, this module allows rotations in both horizontal and vertical dimensions to align with mouse brain samples. To determine the appropriate rotation angles, we manually pinpointed anatomical landmarks corresponding to the samples. For the left-right axis rotation, we chose points from the CA3 pyramidal layer in both hemispheres and the posterior slice of the dentate gyrus, specifically the granule cell layer. For the dorsal-ventral axis rotation, we identified key anatomical landmarks. These include the initial connections of the anterior commissure and corpus callosum between the hemispheres, as well as ventral regions like the interpeduncular nucleus and the suprachiasmatic nucleus. After determining these rotation angles, we adjusted the reference atlas to match our samples.

Image registration module

This module uses a tool that aligns brain slice images with a reference atlas, facilitating the alignment of overall brain region distribution. Registration starts by matching the coronal plane of the sample sections to the atlas. After defining the external boundaries of the brain section, the system performs geometric transformations on the section to optimize its fit with the atlas.

Signal detection module

The detection module is designed to automatically label c-Fos⁺ cells. Post-detection, a manual verification was performed on each digitized brain section image to ensure the accuracy and precision of the c-Fos⁺ cell markings.

Principal components analysis (PCA) of the activity patterns at whole brain

We utilized arithmetic methods to compute a logarithm ratio of the c-Fos density in a specific brain region for PCA analysis. Specifically, the cell density of a single brain region (defined as the number of c-Fos positive cells in a brain region divided by the volume of that region) was divided by the global c-Fos cell density (calculated as the total number of c-Fos positive cells throughout the entire brain divided by the total brain volume). Subsequently, the logarithm of this ratio value was taken. The formula is as follows:

$$x_R = \log 10 \left[\frac{(N_r + 1)}{V_r} / \frac{\sum_1^R (N_r + 1)}{\sum_1^R V_r} \right]$$

Nr: The number of c-Fos+ cells in each brain region. Vr: The volume of each brain region. Principal components analysis was performed on the concatenated matrix containing data from four conditions, using singular value decomposition. We then selected the first two principal components (PCs), which accounted for 56.6% of the variance in [Figure 2A](#) and 44.5% in [Figure 3A](#).

Network generation

To evaluate how functional connectivity changed under general anesthetics in WT mice, we extracted 63 brain regions from major brain subdivisions (cerebral cortex, cerebral nuclei, thalamus, hypothalamus, midbrain, and hindbrain) listed in [Supplementary Table 2](#). Correlation matrices were generated by computing Pearson correlation coefficients from interregional c-Fos expression in the 63 regions. Mean correlations were calculated to assess changes in functional connectivity between these major subdivisions of the brain. Weighted undirected networks were constructed by considering correlations with Pearson's $r \geq 0.82$, corresponding to a one-tailed significance level of $P < 0.05$ (uncorrected). The nodes in the networks represent brain regions, and the correlations that survived thresholding were considered connections. Theoretical graph analysis was performed using Brain Connectivity Toolbox (<https://sites.google.com/site/bctnet/>, version 2019-03-03) in MATLAB R2021 (the MathWorks Inc.) [47]. Network visualization was performed using Cytoscape (version 3.2.1) [48].

Hub identification

Network centrality was evaluated using degree, betweenness, within-module z-scores (representing within-module connectivity), and participation coefficient (indicating between-module connectivity). These measures were computed for all nodes to identify potential hub regions [49]. Modularity was assessed employing Newman's spectral community detection algorithm [50]. Degree represents the number of edges connected to a node, while betweenness denotes the number of shortest paths traversing a given node. Nodes with elevated betweenness centrality are involved in numerous shortest paths.

Statistical analysis

Sample size was determined based on prior studies [13, 14]. Data are presented as mean $\pm$ SEM, and all statistical tests were two-sided. For the c-Fos cell percentages in [Figure 2C](#), we employed the Mann-Whitney test with a false discovery rate correction, setting a Q-value at 0.05. In [Figures 4](#) and [5](#), differences in c-Fos cell counts for both KET and ISO were evaluated using an independent t-test. P-values were adjusted using the procedure by Benjamini, Krieger, and Yekutieli with Q set at 0.05 [51]. [Figure 6C](#) and [Supplementary Figure 1](#) utilized a one-way ANOVA followed by a post-hoc Tukey correction. Pearson correlation coefficients (R) were transformed into Z-scores using the Fisher's Z transformation before computing group means and

making statistical comparisons. All statistical analyses were conducted using GraphPad Prism 9.0 (GraphPad Software, USA) and MATLAB R2021 (Mathworks Inc.). A P-value of less than 0.05 was deemed statistically significant.

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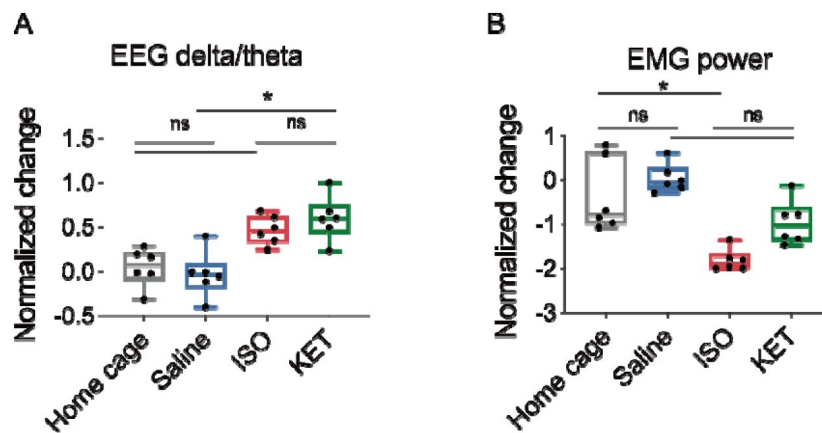
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Author contributions

Yue Hu, Conceptualization, Formal analysis, Investigation, Visualization, Methodology, Writing—original draft, Writing—review and editing; Jiangtao Qi, Conceptualization, Software, Formal analysis, Investigation, Methodology, Writing—original draft, Writing—review and editing; Zhao Zhang, Resources, Investigation, Supervision, Writing—review and editing; Mengqiang Luo, Resources, Supervision, Funding acquisition, Project administration, Writing—review and editing; Yingwei Wang, Conceptualization, Resources, Supervision, Funding acquisition, Project administration, Writing—review and editing.

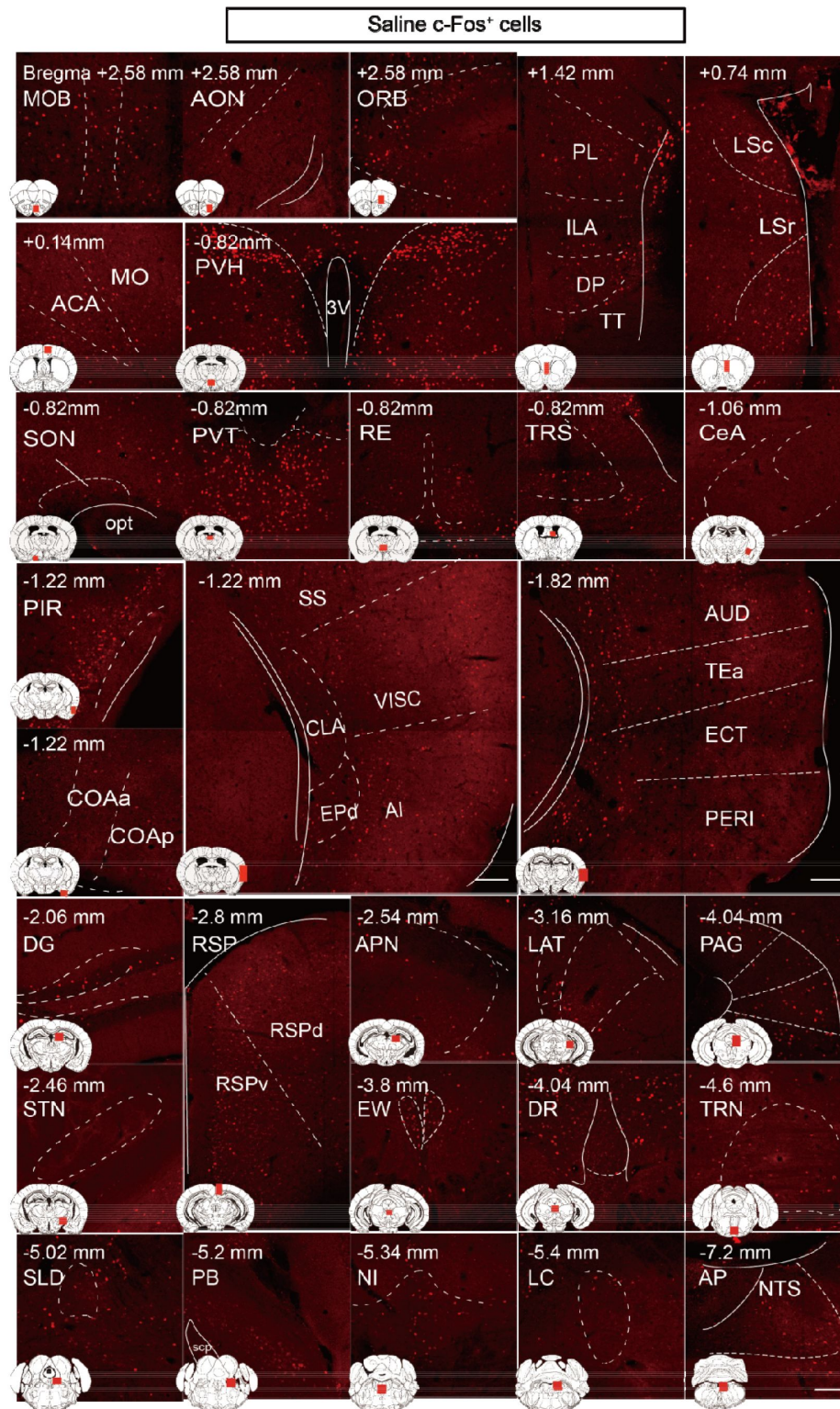
Supplementary Information



Supplementary Figure 1.

EEG and EMG power change after each treatment.

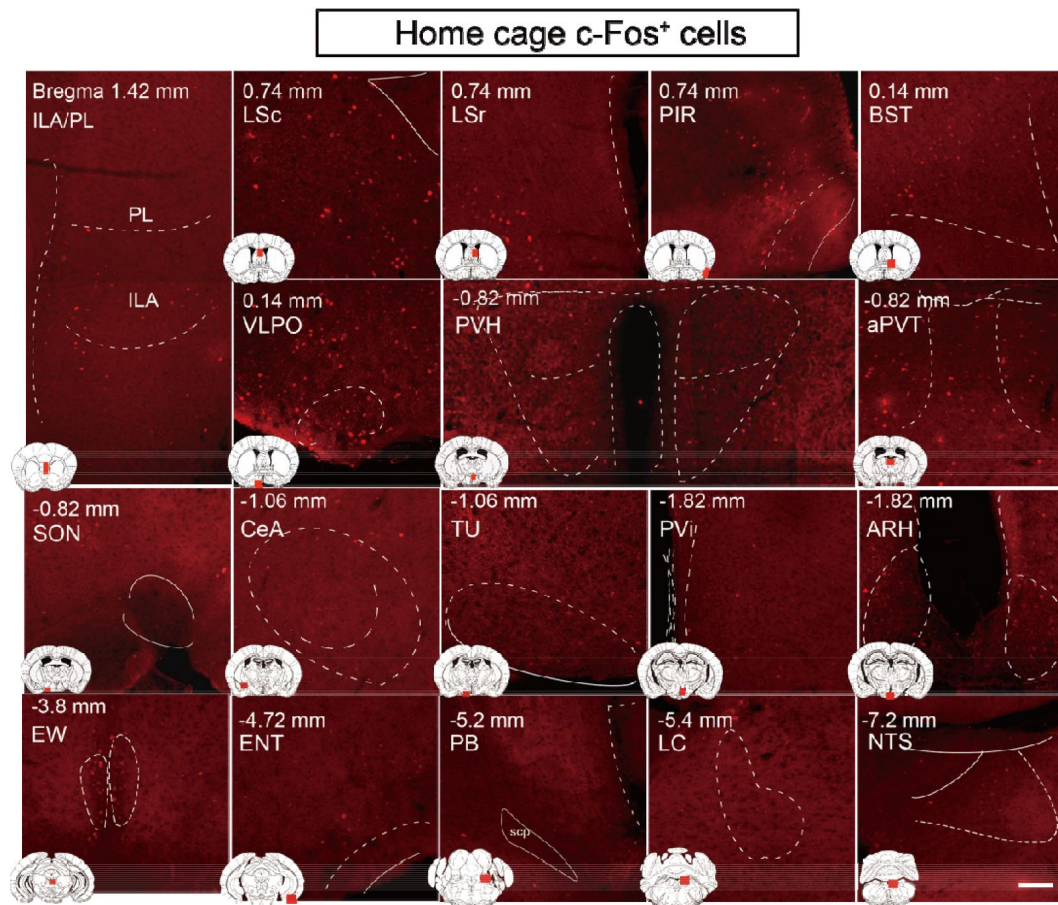
Box represents the 25th-75th percentiles; the central line denotes the median; whiskers indicate maximum and minimum values. $n = 6, 6, 6, 6$ for the home cage, ISO, saline, and KET, respectively. (A) Normalized Change in EEG power: ISO vs KET, $P > 0.99$; Saline vs KET, $P = 0.01$; Home cage vs ISO, $P = 0.11$. (B) Normalized Change in EMG power: ISO vs KET, $P = 0.36$; Saline vs KET, $P = 0.30$; Home cage vs ISO, $P = 0.02$. Analyses were conducted using the Kruskal-Wallis test, followed by Dunn's multiple comparisons test.



Supplementary Figure 2.

c-Fos expression in distinct brain regions after exposure to normal saline administration.

(A) Representative immunohistochemical staining of MOB, AON, ORB, MPO, ACA, MO, TRS, PL, ILA, DP, LS, PVT, SON, PVH, RE, VISC, AI, CLA, EPd, PIR, COA, AUD, TEa, ECT, PERI, CeA, SS, DG, STN, RSP, APN, LAT, EW, DR, PAG, SLD, PB, TRN, NI, LC, and NTS c-Fos⁺ cells from the indicated mice. Scale bar, 200 μ m.



Supplementary Figure 3.

c-Fos expression in home cage group.

(A) Representative immunohistochemical staining of PL, ILA, LSc, LSr, PIR, BST, VLPO, PVH, aPVT, SON, CeA, TU, PVl, ARH, EW, ENT, PB, LC, and NTS c-Fos⁺ cells from the indicated mice. Scale bar, 200 μ m.

Brain Structure	Substructure	Abbreviation	Home cage	ISO	P 1	Saline	KET	P 2
Pallidum	Pallidum,dorsal region	PALd	0.057±	0.010±	0.275	0.080±	0.131±	0.179
			0.028	0.005		0.048	0.065	
	Pallidum,ventral region	PALv	0.253±	0.712±	0.009	0.363±	0.785±	0.020
			0.083	0.093		0.157	0.106	
	Pallidum,medial region	PALm	0.311±	0.575±	0.065	0.305±	0.511±	0.081
			0.096	0.054		0.080	0.060	
	Pallidum,caudal region	PALc	1.108±	3.648±	0.002	0.626±	0.402±	0.852
			0.184	0.356		0.186	0.107	
Striatum	Striatum dorsal region	STRd	0.500±	0.572±	0.937	1.191±	3.466±	0.013
			0.126	0.183		0.543	0.840	
	Striatum ventral region	STRv	0.589±	2.774±	0.002	0.597±	2.312±	0.005
			0.098	0.402		0.267	0.287	
	Lateral septal complex	LSX	1.932±	5.108±	0.004	2.285±	1.052±	0.135
			0.586	0.413		0.610	0.059	
	Striatum-like amygdalar nuclei	sAMY	1.135±	6.541±	0.002	1.830±	0.952±	0.142
			0.376	0.541		0.417	0.283	
Olfactory areas	Main olfactory bulb	MOB	0.677±	1.702±	0.177	1.420±	2.792±	0.217
			0.292	0.517		0.791	1.255	
	Accessory olfactory bulb	AOB	0.109±	0.208±	0.697	0.178±	0.419±	0.225
			0.066	0.139		0.093	0.228	
	Anterior olfactory nucleus	AON	0.999±	1.117±	0.485	1.053±	2.376±	0.041
			0.548	0.354		0.486	0.681	
	Taenia tecta	TT	0.469±	1.193±	0.093	0.463±	0.700±	0.172
			0.099	0.274		0.280	0.254	
	Dorsal peduncular area	DP	0.020±	0.171±	0.002	0.075±	0.162±	0.103
			0.009	0.047		0.035	0.044	
	Piriform area	PIR	1.300±	4.166±	0.093	3.248±	4.515±	0.282
			0.356	1.449		0.528	0.747	

Supplementary Table 1

c-Fos-positive cell counts in each brain region normalized to the total cell count of their respective whole brains for the home Cage, ISO, saline, and KET groups. P1 denotes p-values for comparisons between the Home Cage and ISO groups, while P2 represents p-values for the contrasts between the Saline and KET groups. Statistical evaluations were undertaken using the Mann-Whitney test, followed by a false discovery rate (FDR) correction, with a threshold Q-value established at 0.05.

	Nucleus of the lateral olfactory tract	NLOT	0.097± 0.044	0.091± 0.040	0.699	0.240± 0.120	0.090± 0.029	0.414
	Cortical amygdalar area	COA	0.365± 0.114	0.430± 0.096	0.589	1.140± 0.193	0.878± 0.210	0.662
	Piriform-amygdalar area	PAA	0.063± 0.018	0.120± 0.040	0.329	0.336± 0.076	0.213± 0.049	0.345
	Post piriform transition area	TR	0.324± 0.131	0.130± 0.058	0.387	0.350± 0.089	0.325± 0.072	>0.9999 99
<i>Isocortex</i>	Somatomotor areas	MO	0.736± 0.225	0.263± 0.105	0.093	0.715± 0.225	4.671± 0.244	0.001
	Somatosensory areas	SS	0.775± 0.309	0.231± 0.123	0.093	1.058± 0.489	9.487± 1.551	0.001
	Gustatory areas	GU	0.032± 0.019	0.048± 0.018	0.387	0.040± 0.017	0.288± 0.044	0.001
	Visceral area	VISC	0.014± 0.007	0.004± 0.003	0.318	0.147± 0.071	0.564± 0.142	0.029
	Auditory areas	AUD	0.256± 0.093	0.246± 0.081	>0.9999 99	1.481± 0.427	2.267± 0.198	0.142
	Visual areas	VIS	0.501± 0.247	0.189± 0.044	0.818	4.503± 1.343	7.013± 0.775	0.181
	Anterior cingulate area	ACA	0.183± 0.078	0.146± 0.037	0.981	0.736± 0.337	2.179± 0.145	0.020
	Prelimbic area	PL	0.118± 0.043	0.160± 0.053	0.485	0.283± 0.100	0.649± 0.086	0.028
	Infralimbic area	ILA	0.279± 0.056	0.922± 0.144	0.002	0.409± 0.182	0.615± 0.217	0.393
	Orbital area	ORB	0.337± 0.083	0.393± 0.054	0.485	0.453± 0.167	1.759± 0.152	0.001
	Agranular insular area	AI	0.146± 0.047	0.096± 0.032	0.485	0.533± 0.164	1.475± 0.231	0.013
	Retrosplenial area	RSP	1.007± 0.349	0.552± 0.114	0.394	2.455± 0.721	5.070± 0.773	0.043
	Posterior parietal association areas	PTLp	0.149± 0.032	0.035± 0.016	0.041	0.238± 0.111	0.598± 0.359	0.634
	Temporal association areas	TEa	0.266± 0.104	0.513± 0.194	0.310	1.501± 0.433	1.539± 0.135	0.573
	Ectorhinal area	ECT	0.305± 0.144	0.901± 0.384	0.169	1.507± 0.341	1.301± 0.341	0.755
	Frontal Pole, cerebral cortex	FRP	0.020± 0.013	0.030± 0.027	>0.9999 99	0.074± 0.043	0.029± 0.012	0.750
Hippocamp	Hippocampal	HIP	4.240±	3.531±	0.699	4.281±	2.938±	0.228

Supplementary Table 1 (continued)

al	region		1.230	0.843		0.766	0.469	
formation	Retrohippocampal region	RHP	1.689± 0.500	3.558± 1.068	0.132	4.627± 1.113	9.551± 0.967	0.013
Cortical subplate	Cortical subplate	CTXsp	1.122± 0.216	0.910± 0.181	0.589	2.798± 0.440	2.410± 0.226	0.662
Thalamus	Thalamus, sensory motor cortex-related	DORsm	0.881± 0.160	0.505± 0.185	0.132	0.779± 0.102	0.546± 0.082	0.181
	Thalamus, polymodal association cortex-related	DORpm	7.353± 1.042	6.104± 0.601	0.180	4.674± 0.992	3.502± 0.259	0.662
	Periventricular zone	PVZ	2.330± 0.759	6.627± 0.843	0.015	1.306± 0.345	0.334± 0.068	0.008
	Periventricular region	PVR	9.800± 2.909	8.145± 0.869	0.937	3.140± 0.421	0.762± 0.088	0.001
Hypothalamus	Hypothalamic medial zone	MEZ	6.198± 1.192	3.878± 0.204	0.240	5.121± 0.832	1.349± 0.174	0.001
	Hypothalamic lateral zone	LZ	2.815± 0.304	5.589± 0.752	0.002	2.606± 0.718	0.883± 0.167	0.005
	Midbrain, sensory-related	MBsen	11.525 ±3.354	2.449± 0.735	0.065	7.507± 2.343	1.466± 0.332	0.020
	Midbrain, motor-related	MBmot	10.099 ±1.299	4.515± 1.105	0.041	11.659 ±1.716	3.712± 0.588	0.005
Midbrain	Midbrain, behavioral state-related	MBsta	0.471± 0.125	0.535± 0.145	0.699	0.834± 0.168	0.380± 0.099	0.043
	Pons, sensory-related	P-sen	1.729± 0.917	0.806± 0.261	0.937	0.842± 0.149	0.661± 0.231	0.573
	Pons, motor-related	P-mot	3.579± 1.115	0.715± 0.197	0.015	2.772± 0.562	1.168± 0.247	0.081
	Pons, behavioral state-related	P-sat	0.816± 0.314	0.572± 0.115	0.818	1.286± 0.308	0.564± 0.109	0.059
Medulla	Medulla, sensory-related	MY-sen	0.618± 0.241	2.042± 0.426	0.015	0.700± 0.249	0.356± 0.098	0.345
	Medulla, motor-related	MY-mot	3.410± 1.727	1.542± 0.144	>0.9999 99	1.347± 0.369	1.476± 0.374	0.852
	Medulla, behavioral state-related	MY-sat	0.020± 0.013	0.019± 0.008	0.740	0.024± 0.011	0.078± 0.032	0.043
Cerebellum	Cerebellum	CB	3.451±	1.379±	0.310	1.869±	0.508±	0.020

Supplementary Table 1 (continued)

m	1.469	0.352	0.611	0.245
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Shown are the mean $\pm$ SEM. of the percentage of c-Fos⁺ cells in each area.

Supplementary Table 1 (continued)

Brain regions	Function	Treatment Regimen	Durati on	Time of day	Effect	References
CeA	Pain Suppression	100mg/kg and 10mg/kg xylazine i.p.	2h	/	Up-regulated	Hua T, et al.,(2022) [17]
SON	Natural sleep/General anesthesia	100mg/kg with or without xylazine(10mg/kg) i.p	2h	/	Up-regulated	Jiang-Xie LF, et al.,(2019) [18]
pCC	/	100mg/kg i.p.	2h	/	Up-regulated	Nagata A, et al.,(1998) [53]
pCC, RSC	/	100mg/kg i.p.	2h	/	Up-regulated	Nagata A, et al.,(2001) [54]
RSP, CeA, Midline thalamic nuclei	Antidepressant	10mg/kg i.p.	1.5h	/	Up-regulated	Inta D, et al.,(2009) [55]
Cg, BF, Orx, TMN, vIPAG, LC, LHb, EW, midline thalamus	LC/vPAG engaged in analgesia	150mg/kg, 60mg/kg, 30mg/kg i.p.	2h	ZT3	Up-regulated	Lu J, et al.,(2008) [13]
CC, PVT	/	100mg/kg i.p.	/	/	/	Nakao S, et al.,(1993) [56]
PC, RSP	/	50mg/kg i.p.	2h	ZT3-9	Up-regulated	Nakao S, et al.,(2002) [57]

Abbreviation: CeA, Central amygdalar nucleus; SON, Supraoptic nucleus; pCC, Posterior cingulate cortex; RSC, Retrosplenial cortex; RSP, Retrosplenial area; Cg, cingulate cortex; BF, basal forebrain cholinergic cells; Orx, perifornical orexin cells; vIPAG, ventrolateral periaqueductal gray

Supplementary Table 2.

Sum of previous studies on ketamine activated brain regions detected by c-Fos immunostaining.

dopamine cells; LC, Locus coeruleus; LHb, lateral habenular nucleus; EW, Edinger-Westphal nucleus; CC, Cerebral cortex; PVT, Paraventricular nucleus of the thalamus; PC, posterior cingulate. RSP, Retrosplenial area.

Supplementary Table 2. (continued)

Brain regions	Function	Concentration of ISO	Duration	Time of day		References
VLPO	Sleep-Promoting, Hypnosis	1.2%, 0.6%, 0.3%	2h	ZT12-14/ZT4-6	Up-regulated	Moore JT, et al.,(2012) [7]
PIR, LSD, LSV	/	4% 5mins+1%	2h	ZT2-7	Up-regulated	Smith ML, et al.,(2016) [12]
NTS, VeN	Vomiting	1% and 3%	1.5h	/	Up-regulated	Gupta RG, et al.,(2016) [38]
CeA	Pain Suppression	1.5%	2h	/	Up-regulated	Hua T, et al.,(2022) [17]
SON	Natural sleep, General anesthesia	1%,1.2%	2h	/	Up-regulated	Jiang-Xie LF, et al.,(2019) [18]
VLPO, MnPO	Sleep and unconsciousness	1.2%	2h	ZT14-17	Up-regulated	Han B, et al.,(2022) [15]
EW, VLPO, LC	Sleep-Promoting, analgesia	2%	2h	ZT12	Up-regulated	Lu J, et al.,(2008) [13]
AP	Nausea, vomiting	1.3%,2.6%	2h	/	Up-regulated	Hase T, et al.,(2019) [58]

Abbreviation: VLPO, Ventrolateral preoptic nucleus; PIR, Piriform area; LSD, Lateral septal nucleus, dorsal part; LSV, Lateral septal nucleus, ventral part; NTS, Nucleus of the solitary tract; VeN, Ventral group of the dorsal thalamus; CeA, Central amygdalar nucleus; SON, Supraoptic nucleus; MnPO, median preoptic nucleus; EW, Edinger-Westphal nucleus; LC, Locus coeruleus; AP, Area postrema.

Supplementary Table 3.

The sum of previous studies on isoflurane activated brain regions detected by c-Fos immunostaining.

Region	Abbreviation	Region	Abbreviation	Regio	Abbreviation
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Supplementary Table 4.

Abbreviations of the relevant brain regions in Figure 3 [↗](#).

				n	
CTX	Cerebral cortex	LH	Lateral habenula	OP	Olivary pretectal nucleus
PA	Posterior amygdalar nucleus	SPF	Subparafascicular nucleus	ND	Nucleus of Darkschewitsch
PIR	Piriform area	MH	Medial habenula	VTN	Ventral tegmental nucleus
DG	Dentate gyrus	ATN	Anterior group of the dorsal thalamus	IF	Interfascicular nucleus raphe
ENT	Entorhinal area	VENT	Ventral group of the dorsal thalamus	PPT	Posterior pretectal nucleus
MOB	Main olfactory bulb	LAT	Lateral group of the dorsal thalamus	MRNm	Midbrain reticular nucleus, magnocellular part
CA	Ammon's horn	GENd	Geniculate group, dorsal thalamus	MRNm g	Midbrain reticular nucleus, magnocellular part, general
AON	Anterior olfactory nucleus	RT	Reticular nucleus of the thalamus	MRNp	Midbrain reticular nucleus, parvicellular part
TT	Taenia tecta	SPA	Subparafascicular area	INC	Interstitial nucleus of Cajal
ECT	Ectorhinal area	PP	Peripeduncular nucleus	CS	Superior central nucleus raphe
RSP	Retrosplenial area	PIN	Pineal body	HB Hindbrain	
ILA	Infralimbic area	HY	Hypothalamus	NTS	Nucleus of the solitary tract
SUB	Subiculum	SO	Supraoptic nucleus	PB	Parabrachial nucleus
TEa	Temporal association areas	PVH	Paraventricular hypothalamic nucleus	PG	Pontine gray
COA	Cortical amygdalar area	LHA	Lateral hypothalamic area	VNC	Vestibular nuclei
ORB	Orbital area	MPO	Medial preoptic area	GRN	Gigantocellular reticular nucleus
SS	Somatosensory areas	AHN	Anterior hypothalamic nucleus	LRN	Lateral reticular nucleus
MO	Somatomotor areas	SBPV	Subparaventricular zone	PCG	Pontine central gray
VIS	Visual areas	DMH	Dorsomedial nucleus of the hypothalamus	PRNc	Pontine reticular nucleus, caudal part
PERI	Perirhinal area	TU	Tuberal nucleus	IRN	Intermediate reticular nucleus
AUD	Auditory areas	SCH	Suprachiasmatic nucleus	MDRN	Medullary reticular nucleus
AOB	Accessory olfactory bulb	PH	Posterior hypothalamic nucleus	AP	Area postrema
EP	Endopiriform nucleus	ARH	Arcuate hypothalamic nucleus	PRNr	Pontine reticular nucleus

Supplementary Table 4. (continued)

BMAp	Basomedial amygdalar nucleus, posterior part	LPO	Lateral preoptic area	NLL	Nucleus of the lateral lemniscus
TR	Postpiriform transition area	RCH	Retrochiasmatic area	DMX	Dorsal motor nucleus of the vagus nerve
BLA	Basolateral amygdalar nucleus	VMH	Ventromedial hypothalamic nucleus	TRN	Tegmental reticular nucleus
PL	Prelimbic area	VLPO	Ventrolateral preoptic nucleus	CN	Cochlear nuclei
ACA	Anterior cingulate area	PVpo	Periventricular hypothalamic nucleus, preoptic part	DCN	Dorsal column nuclei
LA	Lateral amygdalar nucleus	MPN	Medial preoptic nucleus	PAS	Parasolitary nucleus
DP	Dorsal peduncular area	PVi	Periventricular hypothalamic nucleus, intermediate part	PGRN	Paragigantocellular reticular nucleus
PAA	Piriform- amygdalar area	ZI	Zona incerta	PARN	Parvicellular reticular nucleus
POST	Postsubiculum	MEPO	Median preoptic nucleus	LC	Locus ceruleus
AI	Agranular insular area	AVPV	Anteroventral periventricular nucleus	SPVI	Spinal nucleus of the trigeminal, interpolar part
BMAa	Basomedial amygdalar nucleus, anterior part	MBO	Mammillary body	PHY	Perihypoglossal nuclei
PAR	Parasubiculum	PVa	Periventricular hypothalamic nucleus, anterior part	NI	Nucleus incertus
NLOT	Nucleus of the lateral olfactory tract	PVHd	Paraventricular hypothalamic nucleus, descending division	LDT	Laterodorsal tegmental nucleus
PRE	Presubiculum	PSTN	Parasubthalamic nucleus	V	Motor nucleus of trigeminal
PTLp	Posterior parietal association areas	PVp	Periventricular hypothalamic nucleus, posterior part	DTN	Dorsal tegmental nucleus
GU	Gustatory areas	ADP	Anterodorsal preoptic nucleus	B	Barrington's nucleus
CLA	Clastrum	PMv	Ventral premammillary	SPVC	Spinal nucleus of the

Supplementary Table 4. (continued)

			nucleus		trigeminal, caudal part
FRP	Frontal pole, cerebral cortex	AVP	Anteroventral preoptic nucleus	MARN	Magnocellular reticular nucleus
IG	Induseum griseum	PMd	Dorsal premammillary nucleus	XII	Hypoglossal nucleus
VISC	Visceral area	PS	Parastrial nucleus	SOC	Superior olivary complex
6b	Layer 6b, isocortex	OV	Vascular organ of the lamina terminalis	SLD	Sublaterodorsal nucleus
FC	Fasciola cinerea	STN	Subthalamic nucleus	AMB	Nucleus ambiguus
CNU	Cerebral nuclei	SFO	Subfornical organ	VII	Facial motor nucleus
BST	Bed nuclei of the stria terminalis	PD	Posterodorsal preoptic nucleus	NTB	Nucleus of the trapezoid body
CEA	Central amygdalar nucleus	PST	Preparasubthalamic nucleus	SPVO	Spinal nucleus of the trigeminal, oral part
LSr	Lateral septal nucleus, rostral (rostroventral) part	ASO	Accessory supraoptic group	RPO	Nucleus raphe pontis
SI	Substantia innominata	AHA	Anterior hypothalamic area	RO	Nucleus raphe obscurus
NDB	Diagonal band nucleus	PSCH	Suprachiasmatic preoptic nucleus	IO	Inferior olivary complex
ACB	Nucleus accumbens	IPN	Interpeduncular nucleus	SUT	Supratrigeminal nucleus
MS	Medial septal nucleus	MB	Midbrain	RM	Nucleus raphe magnus
OT	Olfactory tubercle	ICc	Inferior colliculus, central nucleus	LIN	Linear nucleus of the medulla
LSc	Lateral septal nucleus, caudal (caudodorsal) part	SCsg	Superior colliculus, superficial gray layer	x	Nucleus x
AAA	Anterior amygdalar area	ICe	Inferior colliculus, external nucleus	PSV	Principal sensory nucleus of the trigeminal
FS	Fundus of striatum	SCig	Superior colliculus, motor related, intermediate gray layer	y	Nucleus y
MA	Magnocellular nucleus	ICd	Inferior colliculus, dorsal nucleus	SLC	Subceruleus nucleus
IA	Intercalated amygdalar	SCzo	Superior colliculus, zonal layer	ECU	External cuneate nucleus

Supplementary Table 4. (continued)

	nucleus				
MEA	Medial amygdalar nucleus	SCdg	Superior colliculus, motor related, deep gray layer	VI	Abducens nucleus
CP	Caudoputamen	SCop	Superior colliculus, optic layer	RPA	Nucleus raphe pallidus
SF	Septofimbrial nucleus	EW	Edinger-Westphal nucleus	LTN	Lateral tegmental nucleus
BA	Bed nucleus of the accessory olfactory tract	PRC	Precommissural nucleus	PRNv	Pontine reticular nucleus, ventral part
SH	Septohippocampal nucleus	SCiw	Superior colliculus, motor related, intermediate white layer	SG	Supragenual nucleus
GPe	Globus pallidus, external segment	MPT	Medial pretecal area	SSN	Superior salivatory nucleus
TRS	Triangular nucleus of septum	VTA	Ventral tegmental area	z	Nucleus z
BAC	Bed nucleus of the anterior commissure	DR	Dorsal nucleus raphe	ACVI	Accessory abducens nucleus
GPi	Globus pallidus, internal segment	NPC	Nucleus of the posterior commissure	ACVII	Accessory facial motor nucleus
LSv	Lateral septal nucleus, ventral part	APN	Anterior pretecal nucleus	EV	Efferent vestibular nucleus
PVT	Paraventricular nucleus of the thalamus	SNr	Substantia nigra, reticular part	ECO	Efferent cochlear group
TH	Thalamus	NOT	Nucleus of the optic tract	ICB	Infracerebellar nucleus
RE	Nucleus of reuniens	CLI	Central linear nucleus raphe	ISN	Inferior salivatory nucleus
GENv	Geniculate group, ventral thalamus	RL	Rostral linear nucleus raphe	PMR	Paramedian reticular nucleus
ILM	Intralaminar nuclei of the dorsal thalamus	SNC	Substantia nigra, compact part	PPY	Parapyramidal nucleus
PT	Parataenial nucleus	SCdw	Superior colliculus, motor related, deep white layer	INV	Interstitial nucleus of the vestibular nerve
MED	Medial group of the dorsal thalamus	RN	Red nucleus	SIMmo	Simple lobule, molecular layer

Supplementary Table 4. (continued)

	CTX	Cerebral cortex		HY	Hypothalamus
1	ACA	Anterior cingulate area	32	ARH	Arcuate hypothalamic nucleus
2	AI	Agranular insular area	33	LHA	Lateral hypothalamic area
3	AUD	Auditory areas	34	LPO	Lateral preoptic area
4	CA	Ammon's horn	35	MBO	Mammillary body
5	COA	Cortical amygdalar area	36	MEPO	Median preoptic nucleus
6	DG	Dentate gyrus	37	MPN	Medial preoptic nucleus
7	ECT	Ectorhinal area	38	MPO	Medial preoptic area
8	ENT	Entorhinal area	39	PVH	Paraventricular hypothalamic nucleus
9	ILA	Infralimbic area	40	PVi	Periventricular hypothalamic nucleus, intermediate part
10	MO	Somatomotor areas	41	SO	Supraoptic nucleus
11	MOB	Main olfactory bulb	42	STN	Subthalamic nucleus
12	ORB	Orbital area	43	TU	Tuberal nucleus
13	PIR	Piriform area	44	VLPO	Ventrolateral preoptic nucleus
14	PL	Prelimbic area	45	ZI	Zona incerta
15	RSP	Retrosplenial area		MB	Midbrain
16	SS	Somatosensory areas	46	APN	Anterior pretectal nucleus
17	TEa	Temporal association areas	47	DR	Dorsal nucleus raphe
18	TT	Taenia tecta	48	EW	Edinger-Westphal nucleus
19	VIS	Visual areas	49	PAG	Periaqueductal gray
	CNU	Cerebral nuclei	50	RR	Midbrain reticular nucleus, retrorubral area
20	ACB	Nucleus accumbens	51	SNr	Substantia nigra, reticular part
21	BST	Bed nuclei of the stria terminalis	52	VTa	Ventral tegmental area
22	CEA	Central amygdalar nucleus	53	MRN	Midbrain reticular nucleus
23	LSc	Lateral septal nucleus, caudal (caudodorsal) part	54	PPN	Pedunculo pontine nucleus
24	LSr	Lateral septal nucleus, rostral (rostroventral) part		HB	Hindbrain
25	OT	Olfactory tubercle	55	CS	Superior central nucleus raphe
26	MS	Medial septal nucleus	56	LC	Locus ceruleus
	TH	Thalamus	57	NTS	Nucleus of the solitary tract
27	LAT	Lateral group of the dorsal	58	PB	Parabrachial nucleus

Supplementary Table 5.

The brain regions used for functional network analysis in Figure 6 [↗](#).

		thalamus			
28	LH	Lateral habenula	59	PCG	Pontine central gray
29	PVT	Paraventricular nucleus of the thalamus	60	SLD	Sublaterodorsal nucleus
30	RE	Nucleus of reunions	61	TRN	Tegmental reticular nucleus
31	RT	Reticular nucleus of the thalamus	62	LDT	Laterodorsal tegmental nucleus
			63	PRNr	Pontine reticular nucleus (PRNr)

Supplementary Table 5. (continued)

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Reviewer #1 (Public Review):

Overall, the manuscript has been improved by addressing some of the concerns, however, I am still very confused about the data analysis due to the use of data transformation (relative %fos), the fact that some graphs only show regions that are significant and the interpretation of the PCA analysis which I find inappropriate. Moreover, many answers in the rebuttal did not make it to the final manuscript and are not discussed and limitations raised by the reviewers are not discussed either.

1a. The addition of the EEG/EMG is useful, however, this information is not discussed. For instance, there are differences in EEG/EMG between the two groups (only Ket significantly increased delta/theta power, and only ISO decreased EMG power). These results should be discussed as well as the limitation of not having physiological measures of anesthesia to control for the anesthesia depth.

1b. The possibility that the differences in fos observed may be due to the doses used should be discussed.

1c. The possibility that the differences in fos observed may be due kinetic of anesthetic used should be discussed.

2b. I am confused because Fig 2C seems to show significant decrease in %fos in the hypothalamus, midbrain and cerebellum after KET, while the author responded that " in our analysis, we did not detect regions with significant downregulation when comparing anesthetized mice with controls." Moreover the new figure in the rebuttal in response to reviewer 2 suggests that Ket increases Fos in almost every single region (green vs blue) which is not the conclusion of the paper.

3. There are still critical misinterpretations of the PCA analysis. For instance, it is mentioned that "KET is associated with the activation of cortical regions (as evidenced by positive PC1 coefficients in MOB, AON, MO, ACA, and ORB) and the inhibition of subcortical areas (indicated by negative coefficients) " as well as "KET displays cortical activation and subcortical inhibition, whereas ISO shows a contrasting preference, activating the cerebral nucleus (CNU) and the hypothalamus while inhibiting cortical areas. To reduce inter-individual variability." These interpretations are in complete contradiction with the answer 2b above that there was no region that had decreased Fos by either anesthetic.

4. I still do not understand the rationale for the use of that metric. The use of a % of total Fos makes the data for each region dependent on the data of the other regions which wrongly leads to the conclusion that some regions are inhibited while they are not when looking at the raw data. Moreover, the interdependence of the variable (relative density) may affect the covariance structure which the PCA relies upon. Why not using the PCA on the logarithm of the raw data or on a relative density compared to the control group on a region-per-region basis instead of the whole brain?

Fig. 2B: it's unclear to me why the regions are connected by a line. Such representation is normally used for time series/within-subject series. What is the rationale for the order of the regions and the use of the line? The line connecting randomly organized regions is meaningless and confusing.

Fig 6A. the correlation matrices are difficult to interpret because of the low resolution and arbitrary order of brain regions. I recommend using hierarchical clustering and/or a combination of hierarchical clustering and anatomical organization (e.g. PMID: 31937658). While it is difficult to add the name of the regions on the graph I recommend providing supplementary figures with large high-resolution figures with the name of each brain region so the reader can actually identify the correlation between specific brain regions and the whole brain,

Rationale for Metric Choice: Note that I do not dispute the choice of the log which is appropriate, it is the choice of using the relative density that I am questioning.

5. I am still having difficulties understanding Fig. 3.

Panel A: The lack of identification for the dots in panel A makes it impossible to understand which regions are relevant.

Panel B: what is the metric that the up/down arrow summarizes? Fos density? Relative density? PC1/2?

Panel C: it's unclear to me why the regions are connected by a line. Such representation is normally used for time series/within-subject series. What is the rationale for the order of the regions?

Reviewer #2 (Public Review):

In the revised manuscript, the authors aim to investigate brain-wide activation patterns following administration of the anesthetics ketamine and isoflurane, and conduct comparative analysis of these patterns to understand shared and distinct mechanisms of these two anesthetics. To this end, they perform Fos immunohistochemistry in perfused brain

sections to label active nuclei, use a custom pipeline to register images to the ABA framework and quantify Fos+ nuclei, and perform multiple complementary analyses to compare activation patterns across groups. This is an interesting line of research and a tour de force in brain-wide Fos quantification.

I appreciate many of the changes that were made in the revised manuscript, including FDR correction and transparency in showing their results with and without transformation. However, several key issues described in our first review have not been addressed.

1-Aside from issues with their data transformation (see below), (a) I think they have some interesting Fos counts data in Figures 4B and 5B that indicate shared and distinct activation patterns after KET vs. ISO based anesthesia. These data are far closer to the raw data than PC analyses and need to be described and analyzed in the first figures long before figures with the more abstracted PC analyses. In other words, you need to show the concrete raw data before describing the highly transformed and abstracted PC analyses. (b) This gets to the main point that when selecting brain areas for follow up analyses, these should be chosen based on the concrete Fos counts data, not the highly transformed and abstracted PC analyses.

2-Now, the choice of data transformation for Fos counts is the most significant problem. First, the authors show in the response letter that not using this transformation (region density/brain density) leads to no clustering. However, they also showed the region-densities without transformation (which we appreciate) and it looks like overall Fos levels in the control group Home (ISO) are a magnitude (~10-fold) higher than those in the control group Saline (KET) across all regions shown. This large difference seems unlikely to be due to a biologically driven effect and seems more likely to be due to a technical issue, such as differences in staining or imaging between experiments. Was the Homeage-ISO experiment or at least the Fos labeling and imaging performed at the same time as for the Saline-Ketamine experiment? Please state the answer to this question in the Results section one way or the other.

3-Second, they need to deal with this large difference in overall staining or imaging for these two (Home/ISO and Saline/KET) experiments more directly; their current normalization choice does not really account for the large overall differences in mean values and variability in Fos counts (e.g. due to labeling and imaging differences).

3a-I think one option (not perfect but I think better than the current normalization choice) could be z-scoring each treatment to its respective control. They can analyze these z-scored data first, and then in later figures show PC analyses of these data and assess whether the two treatments separate on PC1/2. And if they don't separate, then they don't separate, and you have to go with these results.

3b-Alternatively, they need to figure out the overall intensity distributions from the different runs (if that the main reason of markedly different counts) and adjust their thresholds for Fos-positive cell detection based on this. I would expect that the saline and HC groups should have similar levels of activation, so they could use these as the 'control' group to determine a Fos-positive intensity threshold that gets applied to the corresponding 'treatment' group.

3c- If neither 3a nor 3b is an option then they need to show the outcomes of their analysis when using the untransformed data in the main figures (the untransformed data plots in their responses to reviewer are currently not in the main or supplementary figs) and discuss these as well.

Reviewer #3 (Public Review):

The present study presents a comprehensive exploration of the distinct impacts of Isoflurane and Ketamine on c-Fos expression throughout the brain. To understand the varying responses across individual brain regions to each anesthetic, the researchers employ principal component analysis (PCA) and c-Fos-based functional network analysis. The methodology employed in this research is both methodical and expansive. Notably, the utilization of a custom software package to align and analyze brain images for c-Fos positive cells stands out as an impressive addition to their approach. This innovative technique enables effective quantification of neural activity and enhances our understanding of how anesthetic drugs influence brain networks as a whole.

The primary novelty of this paper lies in the comparative analysis of two anesthetics, Ketamine and Isoflurane, and their respective impacts on brain-wide c-Fos expression. The study reveals the distinct pathways through which these anesthetics induce loss of consciousness. Ketamine primarily influences the cerebral cortex, while Isoflurane targets subcortical brain regions. This finding highlights the differing mechanisms of action employed by these two anesthetics—a top-down approach for Ketamine and a bottom-up mechanism for Isoflurane. Furthermore, this study uncovers commonly activated brain regions under both anesthetics, advancing our knowledge about the mechanisms underlying general anesthesia.

Author Response

The following is the authors' response to the original reviews.

We sincerely thank the editor and reviewers for their constructive feedback on our manuscript. Based on their recommendations, we've conducted additional experiments, made revisions to the text and figures, and provide a point-by-point response below.

Reviewer #1 (Recommendations for the authors):

- 1. The lack of behavioral/physiological measures of the depth of anesthesia (ventilation, heart rate, blood pressure, temperature, O₂, pain reflexes, etc...) combined with the lack of dose-response and the use of different routes of administration makes the data difficult to interpret. Sure, there is a clear difference in network activation between KET and ISO, but are those effects due to the depth of the anesthesia, the route of administration, and the dose used? The lack of behavioral/physiological measures prevents the identification of brain regions responsible for some of the physiological effects and different effects of anesthetics.*

We greatly appreciate the insightful feedback you have provided.

In response to the concerns about anesthesia depth:

a. We recorded EEG and EMG data both before and after drug administration. Supplementary Figure 1 showcases the changes in EEG and EMG power observed 30 minutes post-drug administration, normalized to a 5-minute baseline taken prior to the drug's administration. Notably, no significant differences were detected in the normalized EEG and EMG power between the ISO and KET groups. Given the marked statistical differences observed between the EEG power in the KET and saline groups, and the EMG power in the home cage and ISO groups, we infer that both anesthetics effectively induced a loss of consciousness.

b. We used standard methods and doses for inducing c-Fos expression with anesthetics, as documented in prior studies (Hua, T, et al., Nat Neurosci, 2020; 23(7): 854-868; Jiang-Xie, L F, et al., Neuron, 2019; 102(5): 1053-1065.e4; Lu, J, et al., J Comp Neurol, 2008; 508(4): 648-62). In future research, it might be more optimal to adopt continuous intraperitoneal or intravenous administration of ketamine.

c. Within the scope of our study, while disparities in anesthesia duration might potentially influence the direct statistical comparison of ISO and KET, such disparities wouldn't compromise the identification of brain regions activated by KET or ISO when assessed as distinct stimuli (ISO vs. home cage; KET vs. saline) or in relation to their individual functional network hub node results.

We hope these additions and clarifications adequately address your concerns and enhance the comprehensibility of our data.

1. Under anesthesia there should be an overall reduction of activity, is that the case? There is no mention of significantly downregulated regions. The authors use multiple transformations of the data to interpret the results (% , PC1 values, logarithm) without much explanation or showing the full raw data in Fig 1. It would be helpful to interpret the data to compare the average fos+ neurons in each region between treatment and control for each drug.

Absence of Significantly Downregulated Regions Under Anesthesia: There are two primary reasons for this observation:

a. Our study's sampling time for the home cage, ISO, saline, and KET groups was during Zeitgeber Time (ZT) 6-7.5. During this period, mice in both the home cage and saline groups typically showed reduced spontaneous activity or were in a sleep state. Our Supplementary Figure 1 EEG and EMG data corroborate this, revealing no significant statistical variations in EEG power between the home cage and ISO groups, nor in EMG power between the saline and KET groups.

b. Our immunohistochemical data showed that the total number of c-Fos positive cells in the two control groups was notably lower than in the experimental groups (Saline group vs KET group: 11808 ± 2386 versus 308705 ± 106131 , $P = 0.006$; Home cage vs ISO group: 3371 ± 840 vs 12326 ± 1879 , $P = 0.001$). This is in line with previous studies, like the one by Cirelli C and team, which found minimal c-Fos expression throughout the mouse brain during physiological sleep (Cirelli, C, and G Tononi, Sleep, 2000; 23(4): 453-69). Thus, in our analysis, we did not detect regions with significant downregulation when comparing anesthetized mice with controls.

Interpreting Raw Data from Figure 1: Regarding the average Fos+ neurons:

In Figures 4 and 5, we utilized raw data (c-Fos cell count) to assess cell expression differences across 201 brain regions within each group. Only brain regions that had significant statistical differences after multiple comparison corrections are shown in the figures.

1. I do not understand their interpretation of the PCA analyses. For instance, in Fig 2 they claim that KET is associated with PC1 while ISO is associated with PC2. Looking at the distribution of points it's clear that the KET animals are all grouped at around +2.5 on PC1 and -2.0 on PC2, this means that KET is associated with both PC1 and PC2 to a similar degree (2 to 2.5). Moreover, I'm confused about why they use PCA to represent the animals/group. PCA is a powerful technique to reduce dimensionality and identify groups of variables that may represent the same underlying construct; however, it is not the best way to identify clusters of individuals or groups.

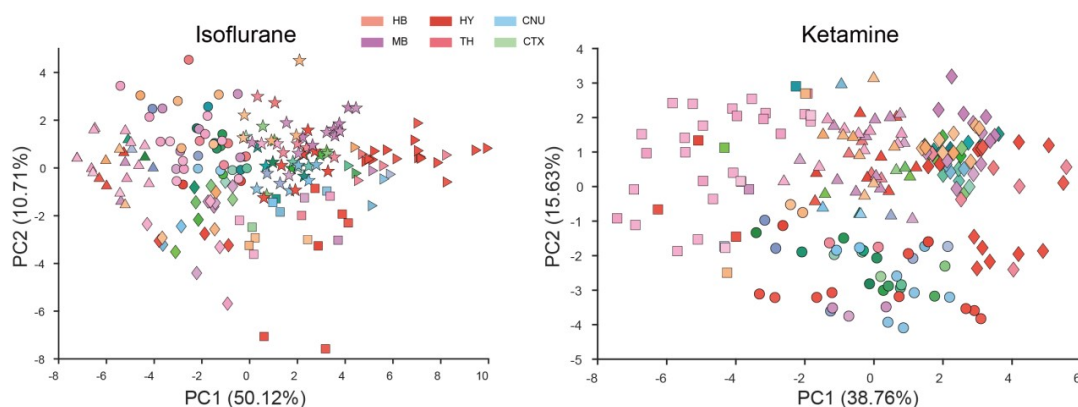
Clarification on PCA Analyses in Figure 2: Thank you for pointing out the ambiguities in our initial presentation of the PCA analyses. We are grateful for the opportunity to address these concerns.

KET and ISO Associations with PC1 and PC2: You rightly observed that KET samples manifest both a positive value on PC1 (around +2.5) and a negative one on PC2 (around -2.0), suggesting that KET has a substantial influence on both principal components. In PCA, a positive score implies a positive association with that component, whereas a negative score suggests a negative association. Contrarily, ISO samples predominantly exhibit values around +2.5 on PC2, with nearly neutral values for PC1, underlining its stronger association with PC2 and lack of significant correlation with PC1. To ensure transparency and clarity, we've adjusted the corresponding descriptions in our manuscript, which can be found on Line 100.

Rationale Behind Using PCA to Represent Animals/Groups: Our initial step was to conduct PCA clustering analysis on the 201 brain regions within both the ISO and KET groups. In the accompanying chart, varying colors denote different brain regions, while distinct shapes represent separate clusters. There wasn't a pronounced distribution pattern within the ISO and KET groups, which led us to adopt the current computational method presented in the paper. This approach was chosen to directly contrast the relative differential expressions between ISO and KET.

We deeply value your feedback, which has steered us toward a clearer and more accurate presentation of our data. We genuinely appreciate your meticulous review.

Author response image 1.



1. The actual metric used for the first PCA is unclear, is it the FOS density in each of the regions (some of those regions are large and consist of many subregions, how does that affect the analysis) is it the %-fos, or normalized cells? The wording describing this is variable causing some confusion. How would looking at these different metrics influence the analysis?

Thank you for raising concerns about the metrics used in our PCA analysis. We recognize the need for clearer exposition and appreciate the opportunity to clarify.

PCA Metrics: The metric for our PCA is calculated by obtaining the ratio of the Fos density within a specific brain region to the global Fos density across the brain. Briefly, this entails dividing the number of Fos-positive cells in a given region by its volume, and then comparing this to the Fos density of the whole brain. The logarithm of this ratio provides our PCA metric. We've elaborated on this in the Materials and Methods section (Lines 401) and enhanced clarity in our revised manuscript, particularly at Line 96.

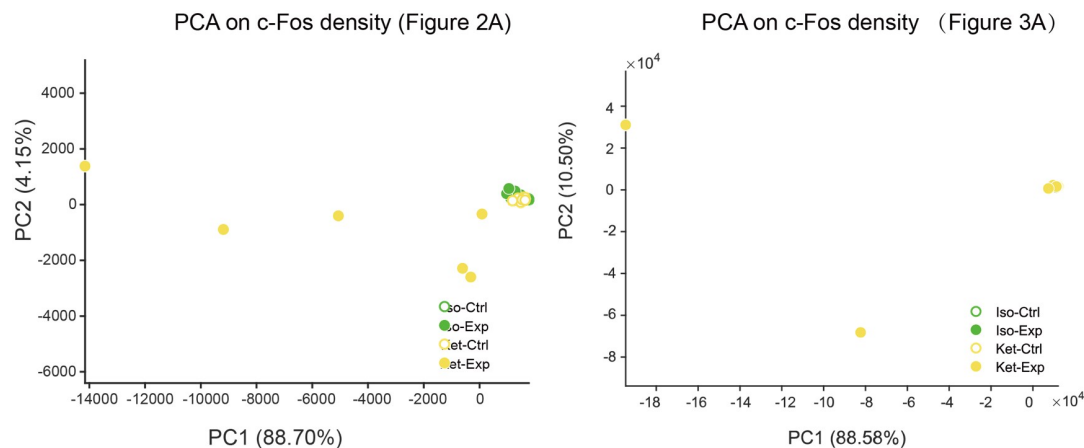
In Figure 2A, we employed 53 larger, mutually exclusive brain regions based on the reference from the study by Do et al. (eLife, 2016;5:e13214). However, in Figure 3A, we used a more detailed segmentation, incorporating 201 distinct brain areas that are more granular than those in Figure 2A. Notably, the PCA results from both representations were consistent. The rationale behind selecting either the 53 or 201 brain regions can be found in our response to Question 10.

Rationale for Metric Choice: The log ratio of regional c-Fos densities relative to the global brain density was chosen due to:

- Notable disparities in c-Fos cell expression across the groups.
- A significant non-normal distribution of density values across animals within the group. Employing the log ratio effectively mitigates the impact of extreme values and outliers, achieving a more standardized data distribution.

We've added PCA plots based on c-Fos densities, depicted in Author response image 2. However, the data dispersion has resulted in a significantly spread-out horizontal scale for these visuals.

Author response image 2.



1. Based on Fig 3 the authors concludes that ISO activates the hypothalamic regions and inhibits the cortex, however, Fig 1 shows neither an activation of the hypothalamus in the ISO nor an inhibition of the cortex when compared to home cage control. If anything it suggests the opposite.

Thank you for your insightful observations regarding the discrepancies between Figures 2 and 3. We believe that when you refer to Figure 1, you are actually referencing Figure 2C.

ISO activation in Hypothalamus: In Figure 2C, we regret the oversight where we inadvertently interchanged the positions of ISO and Saline. When accurately represented, Figure 2C indeed shows that ISO notably activates the periventricular zone (PVZ) and the lateral zone (LZ) of the hypothalamus compared to the home cage group. Moreover, there's a discernible difference in the hypothalamic response between ISO and KET.

ISO's Effect on the Cortex: The main aim of Figure 3 was to highlight the differing responses between ISO and KET in the cortex. Notably, KET demonstrates a positive correlation with PC1 (+7 on PC1), whereas ISO shows a negative association (-3 on PC1). Given that the coefficient of PC1 for the cortical region is positive, it suggests that the cortical areas activated by KET are inhibited by ISO (with KET's distribution around 0 on PC2). However, the divergence between ISO and the home cage is most apparent in PC2, with ISO clusters at +4 and the home cage approximately at -2, suggesting that ISO activates a different set of cortical nuclei. In alignment with this, Figure 2C also illustrates that ISO activates specific cortical areas, such as ILA and PIR, in contrast to the home cage.

Thus, Figure 3 primarily employs PCA to delineate the contrasts between ISO and KET, whereas Figure 2C emphasizes the comparison of each against their respective controls.

1. Control for isoflurane should be air in the induction chamber rather than home cage. It is possible that Fos activation reflects handling/stress pre-anesthesia in the animals, which would increase Fos expression in the stress-related regions such as the BST, striatum (CeA), hypothalamus (PVH) and potentially the LC.

Thank you for emphasizing the importance of an appropriate control for Isoflurane.

In our efforts to minimize the potential impact of stress-induced c-Fos expression, we implemented several precautionary measures. Prior to the experiment, both groups of mice were subjected to handling and acclimatization within the induction chamber over four days. By the day of the experiment, for the mice in the experimental group, we ensured they were comfortable and exhibited no signs of distress or fear—such as cowering or evading. With care, we slowly relocated them to the nearby anesthesia induction chamber. Using 5% ISO, anesthesia was induced promptly, following a meticulously devised protocol to reduce stress impacts on c-Fos expression.

Moreover, existing studies have shown Isoflurane's activation of BST/CeA (Hua, T, et al., Nat Neurosci, 2020, 23: 854-868), PVH (Xu, Z, et al., British Journal of Anaesthesia, 2023, 130: 446-458), and LC (Lu, J, et al., J Comp Neurol, 2008, 508: 648-62), even when using oxygen controls. Such literature supports our findings, indicating that the activation we observed was indeed due to Isoflurane and not purely stress-related.

1. In the Ket network there are a few anticorrelated regions, most of which are amongst the list of the most activated regions, does this mean that the strong correlation results from an overall decreased activation? And if so, is it possible that the ketamine anesthesia was stronger than the isoflurane, causing a more general reduction in activity?

The pronounced correlations observed within the ketamine (KET) network do not signify a generalized decrease in activation. Instead, these correlations reflect significantly enhanced activity in specific regions under KET anesthesia. This amplified correlation is an indication of a more widespread increase in activity, rather than a decrease. These findings are consistent with previous research, which showed that anesthetic doses of ketamine produce patterns of Fos expression in the CNS similar to wakefulness (Lu, J, et al., J Comp Neurol, 2008; 508(4): 648-62).

Regarding the comparative strength of KET versus ISO anesthesia, our electroencephalographic evidence confirms that both agents induce a loss of consciousness. No significant differences were observed in EEG and EMG readings within the first 30 minutes post-administration. In future research, a continuous intravenous or intraperitoneal administration of KET might be a preferable method.

1. Since they have established networks it would be easy and useful to look at how the different regions identified (sleep, pain, neuroendocrine, motor-related, ...) work together to maintain analgesia, are they within the same module? Do they become functionally connected and is this core network of functional connections similar for KET and ISO?

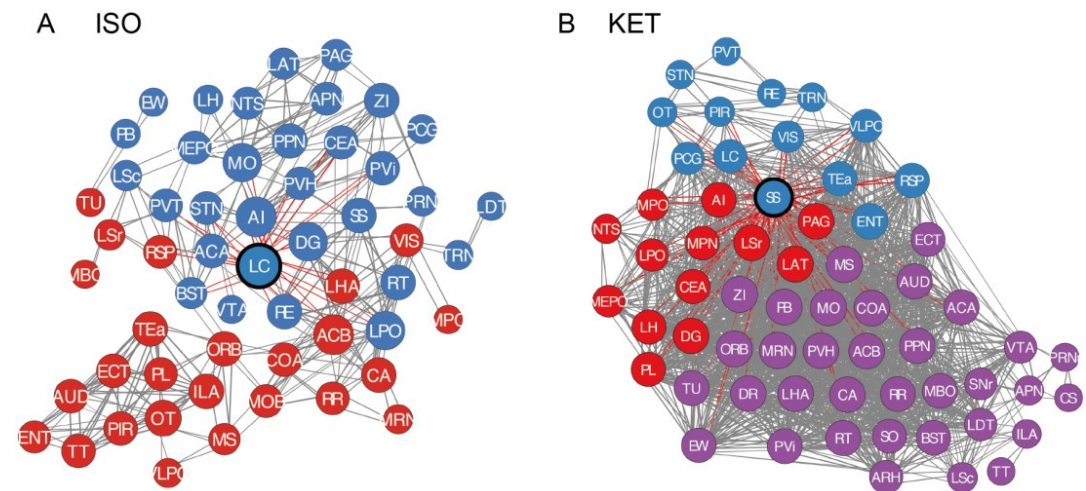
Thank you for your suggestion. In response to your inquiry, we undertook analysis of the core functional networks for KET and ISO, using a set threshold at $r > 0.82$ and $P < 0.05$. For evaluating the modularity of each network, we utilized Newman's spectral community detection algorithm.

(A) The ISO's core functional network (56 nodes, 372 edges) predominantly divides into two modules with a modularity quotient of 0.345. ISO-active regions include arousal-associated regions (PL, ILA, PVT), analgesia-related (CeA, LC, PB), neuroendocrine function nuclei (TU, PVi, ARH, PVH, SON) as detailed in Figure 5. Notably, ARH and SON weren't incorporated into the core network. Analgesia-associated regions, such as CeA, LC, and PB, reside within module 1, while neuroendocrine nuclei are spread between modules 1 and 2.

(B) In contrast, KET's core functional network (61 nodes, 1820 edges) splits into three distinct modules, but its low modularity quotient (0.06) indicates a lack of clear functional modularization, suggesting denser interconnections among brain regions. Furthermore, functionally-related regions such as arousal (PL, ILA, PVT, DR), analgesia-related (ACA, APN, PAG, LC), and neuroendocrine regulation (PVH, SON), etc., as seen in Figure 4, are distributed across different modules. This distribution may implies that functions like analgesia and neuroendocrine regulation are not governed by simple, linear processes, but arise from complex, overlapping pathways spanning various modules and functional zones.

In summary, the core functional networks of ISO and KET differ, with functionally-related regions spanning multiple modules, reflecting their diverse roles in varied physiological regulations.

Author response image 3.



1. The naming of the function of some of the regions is very much debatable. For instance, PL/ILA are named "sleep-wakefulness regulation" regions in the paper. I can think of many more important functions of the PL/IL including executive functions, behavioral flexibility, and emotional control. It is unclear how the functions of all the regions were attributed. I am not sure that this biased labeling of structure-function is useful to the reports, it may instead suggest wrong conclusions.

Thank you for your thoughtful feedback regarding our classification of the functions of the PL/ILA regions in our manuscript.

We recognize the challenge in accurately defining the functions of brain regions. While there is evidence highlighting the role of PL/ILA in arousal pathways, we also acknowledge their documented roles in executive functions, behavioral flexibility, and emotional control. In response to your comments, we have refined our description, changing "sleep-wakefulness regulation" to "wake-promoting pathways" (see Line: 159, 164).

It's worth noting that many brain regions, including the PL/ILA, have multiple functions. We agree that a single label might not capture the entirety of their roles. To provide a broader perspective, we will add a section in our manuscript that sheds light on the varied functions of these regions (Line: 181).

1. A point of concern and confusion is the number of brain regions analyzed. In the introduction, it is mentioned that 987 brain regions are considered, but this is reduced to 53 selected brain regions in Figure 2, then 201 brain regions in Figure 3, and reduced again to 63 for the network analysis. The rationale for selecting different brain regions is not clear.

For the 987 brain regions: Using the standard mouse atlas available at <http://atlas.brain-map.org/>, the mouse brain is organized into nine levels. The broadest category is the grey matter, which then progresses to more specific subdivisions, totaling 987 unique regions.

For the 53 brain regions: To effectively understand the activation patterns of ISO and KET, we started with a broad approach, looking at larger brain areas like the thalamus and hypothalamus. This broad view, presented in Figure 2, focuses on the 5th-level brain regions, encompassing 53 primary areas. This methodology is also employed in the study by Do et al.

(Elife, 2016; 5: e13214). We have added the rationale for selecting these brain regions in the main text (Line: 92).

Regarding the 201 brain regions in Figures 3, 4, and 5: We delved deeper, examining the 6th-level brain regions, a common granularity in neuroscience research. This detailed view allowed us to highlight specific areas, like the CeA and PVH (Line:129).

Finally, for Figures 6 and 7, we selected 63 regions that were activated by both ISO and KET, as well as regions previously reported to be related to the mechanism of general anesthesia (Leung, L, et al., Progress in neurobiology, 2014; 122: 24-44) (Line: 220). Using these regions, we analyzed the correlation of c-Fos expression, aiming to construct a functional brain network with strong positive connections.

We hope this clarifies our approach and the rationale behind our region selection at each stage of the study. Thank you for your attention to this detail.

1. *The statistical analysis does not seem appropriate considering the high number of comparisons. They use simple t-tests without correction for multiple comparisons.*

Thank you for pointing out the concern regarding our statistical analysis. In the revised manuscript, we addressed the issue of multiple comparisons correction in our t-tests. We adopted the statistical methods detailed in the papers by Renier, N, et al., Cell, 2016; and Benjamini, Y, and Y Hochberg, 1995. P-values were adjusted for multiple comparisons using the two-stage linear step-up procedure of Benjamini, Krieger, and Yekutieli, with a false discovery rate (FDR) threshold (Q) of 0.05. This approach is now explained in the Materials and Methods section (Line: 434). After this adjustment, the brain regions we initially identified remained statistically significant. Furthermore, we revisited the original immunohistochemical images to confirm the differences in c-Fos cell expression between the experimental and control groups, reinforcing our conclusions.

1. *There is no statistical analysis in Fig 2C.*

Thank you for bringing to our attention the lack of statistical analysis in Fig 2C. We have now added the relevant statistical data in Supplementary Table 1 and provided annotations in Fig 2C to reflect this.

Reviewer #2

1. *The authors report 987 brain regions in the introduction, but I cannot find any analysis that incorporates these or even which regions they are. Very little rationale is provided for the regions included in any of the analyses and numbers range from 53 in Figure 1, to 201 in Figure 3, to 63 in Figure 6. It would help if the authors could first survey Fos+ counts across all regions to identify a subset that is of interest (significantly changed by either condition compared to control) for follow up analysis.*

Thank you for your insightful comments on the number of brain regions analyzed in our study.

987 Brain Regions: The reference to 987 brain regions from the standard mouse atlas (<http://atlas.brain-map.org/>) represents the entire categorization of the mouse brain across nine levels. We recognize that a comprehensive analysis of all these regions would be valuable, but to ensure clarity and depth, we took a focused approach.

Region Selection Rationale:

Figure 2: Concentrated on 5th-level brain regions (53 areas), inspired by methods from Do et al. (eLife, 2016;5:e13214). This provided a broad overview of c-Fos expression differences. Figures 4 and 5: Delved into 6th-level brain regions (201 areas), a common practice in neuroscience for more detailed study. Figure 6: We focused on 63 regions, which encompass not only the regions activated by both ISO and KET but also those previously reported to be associated with the mechanisms of general anesthesia. Methodological Approach: Our region selection was rooted in identifying areas with significant changes under anesthetic conditions compared to controls. This staged approach allowed a targeted analysis of the most affected regions, ensuring robust conclusions.

Enhancements: We've incorporated comparative analyses of activated brain regions at different hierarchical levels in Figures 4 and 5. For clearer comprehension, we've added clarifications in the manuscript at Lines: 92, 130, and 220.

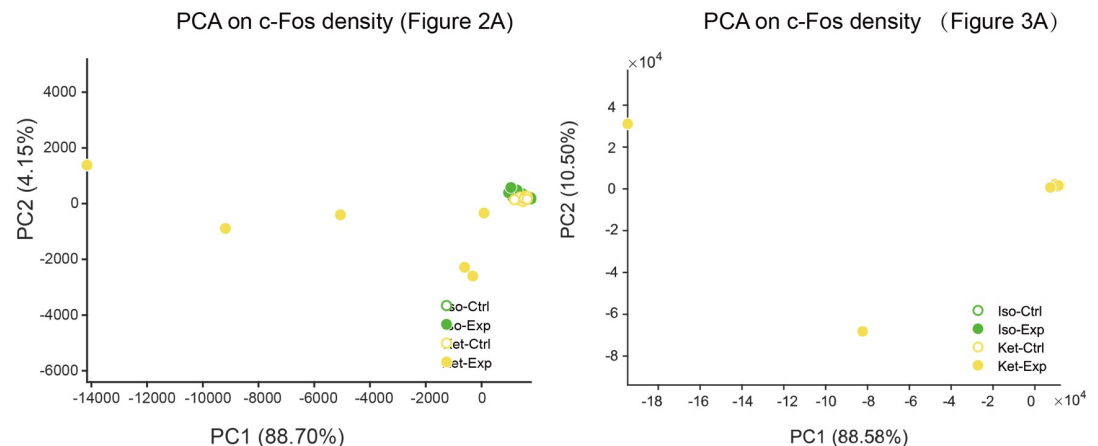
1. Different data transformations are used for each analysis. One that is especially confusing is the 'normalization' of brain regions by % of total brain activation for each animal prior to PCA analysis in Figures 2 and 3. This would obscure any global differences in activation and make it unlikely to observe decreases in activation (which I think is likely here) that could be identified using the Fos+ counts after normalizing for region size (ie. Fos+ count / mm3) which is standard practice in such Fos-based activity mapping studies. While PCA can be powerful approach to identify global patterns, the purpose of the analysis in its current form is unclear. It would be more meaningful to show that regional activation patterns (measured as counts/mm3) are on separate PCs by group.

Thank you for your thoughtful comments. We regret any confusion caused by our initial presentation. For the PCA analysis in Figures 2A and 3A, we calculated the ratio of cell density in each brain region to the overall brain density, and then applied a logarithmic transformation to this ratio. Our approach in Figure 2C was to use the proportion of c-Fos cell counts in individual brain regions to the total cell counts throughout the brain. This methodology considers variations in overall c-Fos cell counts across animals, effectively mitigating potential biases due to differential global activation levels across subjects.

Furthermore, our direct comparison of differences in c-Fos cell counts between ISO, KET, and their respective control groups in Figures 4 and 5 addresses your concerns about potential decreases in activation. Notably, we did not identify any brain regions with significant suppression in these figures, which is consistent with the trends observed post-normalization in Figure 2C.

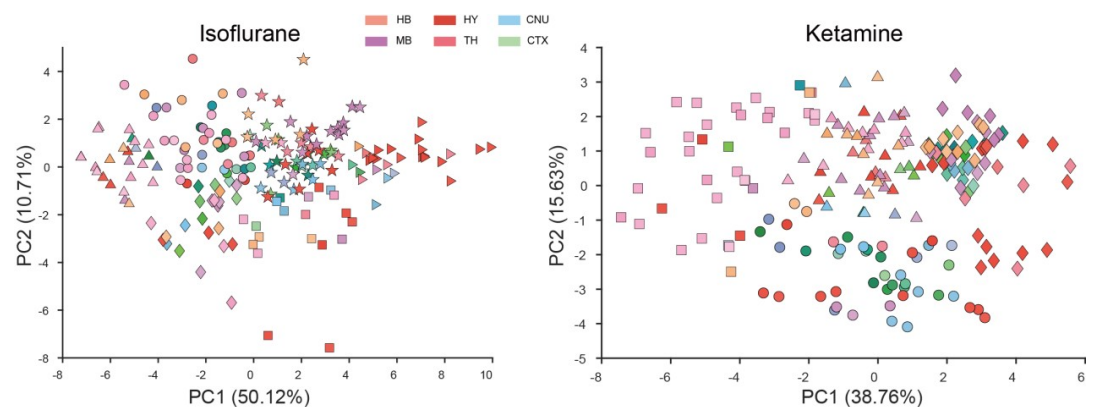
Given your feedback, we conducted another PCA using cell densities for each region (counts/mm3). However, we found significant variability and non-normal distribution of c-Fos density across the groups, leading to extensive data dispersion. Consequently, normalizing the cell counts across regions and then applying a logarithmic transformation before PCA might be more appropriate.

Author response image 4.



Additionally, our exploration of regional activation patterns using PCA analysis for ISO and KET separately, based on the logarithm ratio of the c-Fos density, revealed that there was no distinct clustering feature among the different brain regions (as illustrated in Author response image 5: colors represented distinct brain regions, while the shapes were indicative of different clusters). This observation further suggests that our original statistical approach might be more suitable.

Author response image 5.



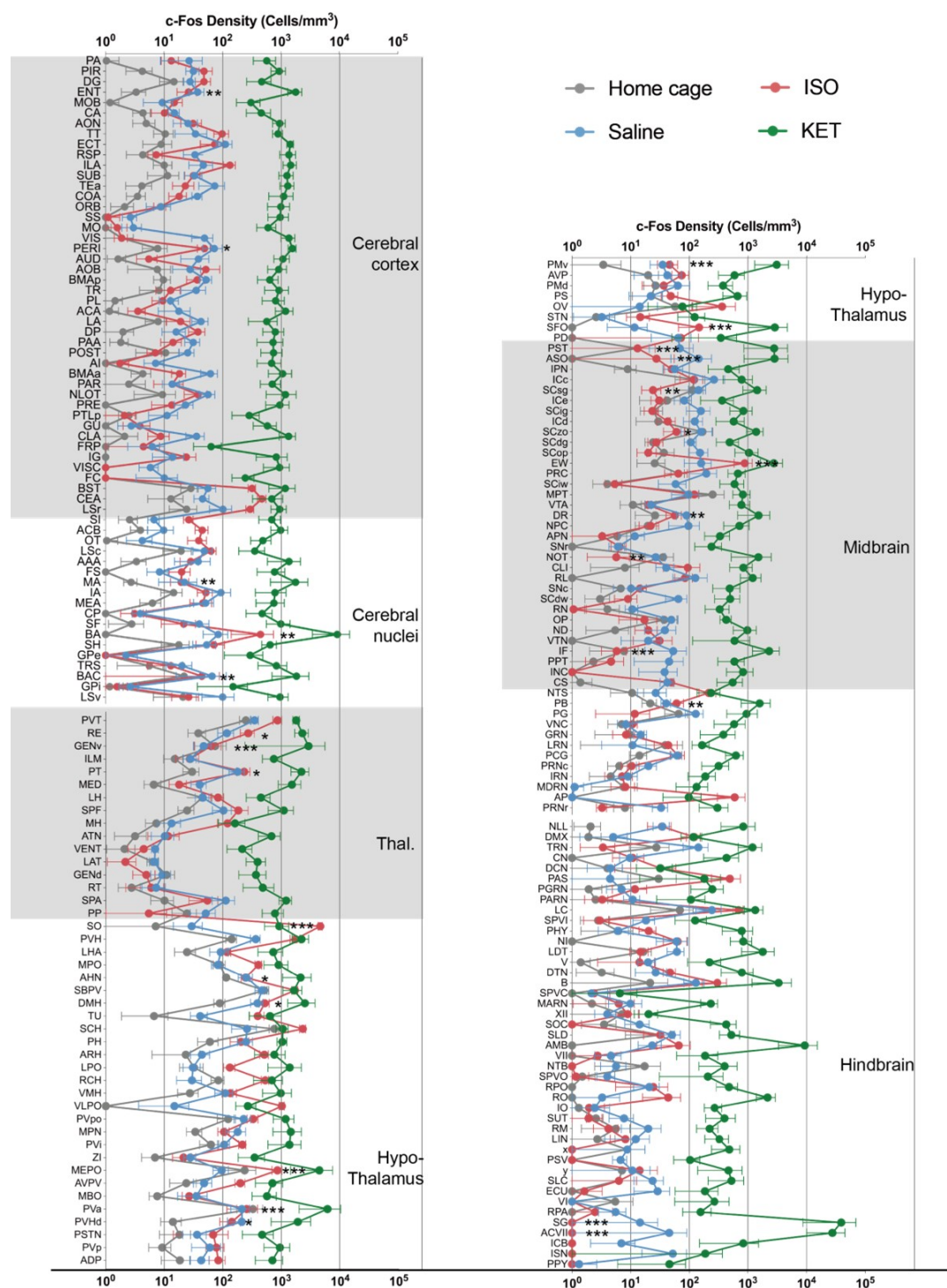
1. Critical problem: The authors include a control group for each anesthetic (ketamine vs. saline, isoflurane vs. homecage) but most analyses do not make use of the control groups or directly compare Fos+ counts across the groups. Strictly speaking, they should have compared relative levels of induction by ketamine versus induction by isoflurane using ANOVAs. Instead, each type of induction was separate from the other. This does not account for increased variability in the ketamine versus isoflurane groups. There is no mention in the Statistics section or in Results section that any multiple comparison corrections were used. It appears that the authors only used Students t-test for each region and did not perform any corrections.

We appreciate the reviewer's insights and have addressed your concerns:

Given the pronounced difference in c-Fos cell count expression between the KET and ISO groups, a direct comparison of Fos+ counts may not effectively capture their inherent

disparities. To better highlight these distinctions, we used the logarithm ratio of c-Fos density in our PCA analysis (Figure 3), mitigating potential disparities in overall cell counts between samples and emphasizing relative variations. However, in response to your feedback, we've included additional analyses. Author response image 6 depicts the c-Fos density (cells/mm³) across different brain regions for the home cage, ISO, saline, and KET groups, with regions like the cerebral cortex, cerebral nuclei, thalamus, and others differentiated by shaded backgrounds. Data are represented as mean $\pm$ SEM. We performed a one-way ANOVA followed by Tukey's post hoc test, marking significant differences between ISO and KET with asterisks: ***P < 0.001, **P < 0.01, *P < 0.05.

Regarding multiple comparison corrections, we've conducted thorough analyses on the data in Figure 2C and Figures 4, 5, and 6, implementing multiple comparison corrections. The detailed methodology is provided in the "Statistical analysis" section.



1. Figures 4 and 5 show brain regions 'significantly activated' following KET or ISO respectively, but again a subset of regions are shown and the stats seem to be t-tests with no multiple comparisons correction. It would help to show these two figures side by side, include the same regions, and keep the y axis ranges similar so the reader can easily compare the 'activation patterns' across the two treatments. Indeed, it looks like KET/Saline induced activation is an order or magnitude or two higher than ISO/Homecage. I would also recommend that this be the first data figure before any other analyses and maybe further analysis could be restricted to regions that are significantly changed in following KET or ISO here.

Thank you for your constructive feedback regarding Figures 4 and 5.

Comparison and Presentation of Figures 4 and 5: We acknowledge your suggestion to present these figures side by side for easier comparison. In the supplementary figure provided in the previous question, we've placed Figures 4 and 5 adjacent to each other, with consistent y-axis ranges, ensuring that readers can make direct comparisons between the activation patterns elicited by KET and ISO.

Statistical Concerns and Region Selection: As mentioned in our previous response, we have conducted multiple comparison corrections on the data presented in Figures 4 and 5. Detailed procedures are elaborated in the "Statistical analysis" section. We believe this approach addresses your concerns regarding the use of t-tests without corrections for multiple comparisons.

Difference in Activation Levels: We observed that the c-Fos activation due to KET is significantly higher than that from ISO. When presented side-by-side using the same scale, ISO activations appear less prominent, potentially mask subtle differences in the activation patterns of ISO, particularly if both KET and ISO showed changes in the same direction in certain brain regions but differed in magnitude. To address this, we used the proportion of c-Fos cell counts in Figure 2C, the logarithm ratio of c-Fos density in Figure 2A and Figure 3. This method emphasizes the relative changes, rather than absolute values, giving a more balanced view of the effects of each treatment.

1. Analyses in Figure 6 and 7 are interesting but again the choice of regions to include is unclear and makes interpreting the results impossible. For example, in Figure 7 it is unclear why the list of regions in bar graphs showing Degree and Betweenness Centrality are not the same even within a single row?

Thank you for your pertinent observation. The choice of brain regions in Figures 6 and 7 was carefully determined based on two main criteria: regions that were significantly activated by ISO or KET within the scope of our study, and those previously reported to be associated with anesthesia mechanisms and sleep-wake regulation.

Regarding your second concern on Figure 7, the discrepancies observed in the x-axes of the bar graphs arise from our methodological approach. We prioritized presenting the top 20% of regions based on their Degree or Betweenness Centrality values. By separately ranking these regions from highest to lowest, the regions presented for each metric inherently differ. This approach was taken to elucidate nodes that consistently emerge as significant across both metrics, thereby highlighting core nodes in the functional network. Were we to use a consistent x-axis without this ranking, it would not only necessitate a more extensive presentation but might also dilute the emphasis on key information. To clarify this methodology and its rationale for our readers, we have expanded upon this in the manuscript at Line 243.

We hope these clarifications address your concerns and facilitate a clearer understanding of our findings.

Reviewer #1 (Recommendations For The Authors):

Minor points

1. In Table 1: the separation of which substructures belong to which brain structure is not clear

1. Line 132 on page 3 seems to repeat the sentence earlier in the paragraph "KET predominantly affects brain regions within the cerebral cortex (CTX), while significantly inhibiting the hypothalamus, midbrain, and hindbrain."

1. Typos

a) Line 99/100 and 130 Central nucleus (CNU) should be cerebral nucleus

b) Comma on line 166

c) Fig. 4D: KET instead of Keta

d) Line 263 "ep"

e) Line 332: 35" "ml (add space)

1. Will data and code be made available?

Thank you for your detailed feedback.

1. We have revised Table 1 to clarify which substructures belong to which brain structures.

2. We acknowledge the redundancy and have now edited line 139 on page 3 to remove the repeated sentence regarding the effects of KET on brain regions.

3. We have addressed the typos you pointed out:

a. The terms "Central nucleus (CNU)" have been corrected to "cerebral nucleus."

b. The comma issue on line 166 has been rectified.

c. In Fig. 4D, we have corrected "Keta" to "KET."

d. We have corrected the typo "ep" on line 263.

e. A space has been added between "35" and "ml" on line 332 as you indicated.

1. Regarding the availability of data and code, we are currently conducting additional analyses related to this study. Once these analyses are completed, we will be more than happy to make the data and code available.

Thank you for assisting us in improving our manuscript.

Reviewer #2 (Recommendations For The Authors):

Minor comments:

1. The term 'whole-brain mapping' in the title suggests that the mapping was performed on 'intact brains' where in fact serial sections were used here. Maybe the authors could change to 'brain-wide mapping' to align better with the study.

Thank you for your insightful comments.

We have revised the title as suggested, changing "whole-brain mapping" to "brain-wide mapping".

1. It is unclear if the mice were kept under anesthesia for the 90-min duration and how the authors monitored the level of sedation. Additionally, if the KET mice were already sedated why were they further sedated with ISO before perfusions and tissue extraction? The methods should be clarified and any potential confounds discussed.

To maintain consistency in the experimental protocol and to reduce stress reactions in the mice, ISO was used before perfusion in all cases. However, this does not affect c-Fos expression as the expression of c-Fos protein starts 20-30 minutes after stimulation (Lara Aparicio, S Y, et al., NeuroSci, 2022; 3(4): 687-702).

We appreciate your guidance in enhancing the clarity of our manuscript.

Reviewer #3 (Recommendations For The Authors):

Recommendation: Minor corrections.

1. The authors should delve deeper into the molecular mechanisms underlying the observed effects, particularly the changes associated with NMDA and GABA receptors. Exploring these mechanisms would provide a more comprehensive understanding of how Ketamine and Isoflurane modulate neural activity and induce anesthesia.
1. The clinical relevance of these findings has not been sufficiently addressed. It would be valuable to elaborate on how the current research outcomes could potentially lead to changes in current anesthesia practices. For instance, identifying the distinct pathways of action for Ketamine and Isoflurane could aid anesthesiologists in selecting the most appropriate anesthetic based on the specific needs of individual patients or surgical procedures.
1. Both Ketamine and Isoflurane have been associated with neurotoxicity. It is important to discuss how the c-Fos activation induced by these anesthetics could contribute, at least partially, to anesthesia-related neurotoxicity. Examining the potential neurotoxic effects would provide a more comprehensive understanding of the risks associated with these anesthetics and aid in the development of safer anesthesia protocols.

Thank you for your valuable suggestions.

Regarding the three points (1, 2, and 3) you've raised, we fully recognize their significance. In the current study, our primary focus was on the differential impacts of Isoflurane and Ketamine on widespread c-Fos expression in the brain. However, we indeed acknowledge the

importance of delving deeper into these mechanisms and their clinical relevance. Therefore, we intend to explore these critical issues in greater detail in our future research endeavors.

We appreciate your feedback, which provides constructive guidance for our subsequent research directions.